Cube-Sum Permutations: Global Involutions, Reflection Entropy, and Algebraic Families

Qichao Wang
Independent mathematical and computational review requested

October 18, 2026

Abstract

Let $P(n)$ count permutations p of $[1, n]$ for which every $i + p(i)$ is an integer cube. We prove that every odd $n \geq 70001$ admits a legal involution with exactly one fixed point, while every even $n \geq 230000$ admits one with no fixed points. These parity-minimal involutions have uniform cube-root entropy:

$$\liminf_{n\to\infty} \frac{\log J_{\min}(n)}{(2n)^{1/3}} \geq \frac{\log 2}{4}.$$

The proof uses two short explicit seed permutations, finite parity-specific odd-cube reflection closures, and a uniform gap induction, with no bipartite matching search. We also establish exponential numbers of parity-minimal involutions along explicit infinite subsequences using taxicab equal-cube identities. We develop further global, threshold-free structure: the cube-sum graph admits a partial matching of size $n - O(n^{2/3})$ for every n, and all Hall obstructions can be restricted to independent sets spanning the two boundary regions. The undirected cycle-cover expansion gives a parity determinant over $\mathbb{F}_2$, while a single valid matching factors exact counting through the strongly connected components of its alternation graph. An explicit four-reflection permutation on 171 elements yields the affine renormalization

$$P(343n + 171) \geq P(n)^{343},$$

and an entropy monotonicity theorem along its multiplicative orbits. Separately, an elementary weighted identity of cubes gives a fully algebraic exponential-subsequence construction

$$P(681707u^3) \geq 2^{148877u^3-1} \qquad (u \geq 1).$$

As a complementary finite classification, 74 exceptional endpoint-multiset identities and three symbolic block families certify the all-n existence conjecture recorded in OEIS A006063, while 191 short augmenting words provide an independent older proof; Hall and alternating-cycle certificates classify the related multiplicity sequence A096680. This last classification remains computer-assisted, unlike the global structural and algebraic constructions. All certificates and independent exact checkers accompany the manuscript; literature priority and external mathematical review remain open.

1 The reflection principle

Write $\mathcal{C} = \{k^3 : k = 2, 3, \dots\}$ and let $P(n)$ be the number of cube-sum permutations of $\{1, \dots, n\}$. By convention the empty permutation is permitted only in the general extension lemma; our classification concerns $n \geq 1$. Let $\mathcal{G} = \{n \geq 1 : P(n) > 0\}$.

Lemma 1 (Reflection and counting)**.** *If $0 \leq m < n$ and $c = m + n + 1$ is a cube, every valid permutation p on $\{1, \dots, m\}$ extends to a valid permutation q on $\{1, \dots, n\}$ via*

$$q_i = \begin{cases} p_i, & i \leq m, \\ c - i, & m < i \leq n. \end{cases}$$

For different admissible cubes c, the images of these extensions are disjoint. In particular,

$$P(n) \geq \sum_{\substack{c\in\mathcal{C} \\ n+1\leq c\leq 2n}} P(c - n - 1), \tag{1}$$

with the convention $P(0) = 1$.

Proof. Because $c = m+n+1$, reflection $i \mapsto c-i$ permutes precisely the new interval $\{m+1, \dots, n\}$, and $i+q_i = c$ there. Old indices retain their cube sums. The embedding is injective for each c and two different c yield different last entries $q_n = c - n$. The claimed sum counts disjoint families. □

Lemma 2 (The cube-gap bound)**.** *For each $n \geq 365$ some cube c satisfies $n + 174 \leq c \leq 2n$. For each $n \geq 556$ some cube c satisfies $n + 176 \leq c \leq 2n$.*

Proof. For $365 \leq n \leq 555$, take $c = 729$. For $n \geq 556$, let k be the least integer with $k^3 \geq n + 176$; then $k \geq 10$ and $n \geq (k-1)^3 - 175$. Thus

$$2n - k^3 \geq k^3 - 6k^2 + 6k - 352 \geq 0.$$

The final polynomial equals 108 at $k = 10$ and has strictly positive successive differences $3k^2 - 9k + 1$ for $k \geq 10$. Hence the second assertion holds and implies the first for $n \geq 556$. □

2 Global graph structure, factorization, and parity

The cube-sum condition is symmetric in its two indices. This observation gives a conceptual interpretation of existence, multiplicity, exact counting, and parity that does not refer to any numerical threshold.

Let G_n be the undirected graph on $[1, n]$, with an edge $\{i, j\}$ whenever $i \neq j$ and $i + j$ is a perfect cube. At a vertex i, allow a loop exactly when $2i$ is a cube. A *spanning cycle-cover configuration* is a decomposition of its vertices into disjoint allowed loops, single edges (dimer components), and unoriented simple cycles of length at least three.

Lemma 3 (Arithmetic location of fixed points)**.** *In any cube-sum permutation, a fixed point $p(i) = i$ can occur only at*

$$\boxed{i = 4r^3 \qquad (r \geq 1).}$$

Consequently a valid involution of odd length must have at least one fixed point at such an index. In particular, a cube-sum permutation of odd length $n < 4$ cannot be an involution.

Proof. At a fixed point $2i = k^3$. The cube on the right is even, so $k = 2r$, whence $i = 4r^3$. An involution partitions its vertices into fixed points and transpositions; when n is odd at least one fixed point is necessary. □

Theorem 1 (Independent-set Hall criterion)**.** *A cube-sum permutation exists on $[1, n]$ if and only if every vertex set I of G_n containing no internal edge* or loop *satisfies*

$$\boxed{|N_{G_n}(I)| \geq |I|,} \tag{2}$$

where $N_{G_n}(I)$ includes all vertices adjacent to some vertex of I. Thus the full family of Hall inequalities can be reduced to independent *vertex sets in the symmetric cube-sum graph.*

Proof. The bipartite double cover of G_n has a perfect matching precisely when Hall's inequality $|N(S)| \geq |S|$ holds for every $S \subseteq [1, n]$. Its perfect matchings are exactly the cube-sum permutations. Hence the proposed condition is necessary.

Conversely, suppose Hall's inequality fails for some S. Put $I = S \setminus N(S)$. No two vertices of I can be adjacent, nor can I contain a loop: otherwise one of those vertices would lie in $N(S)$. Moreover $N(I) \cap S = \varnothing$, because any edge from I to S would again contradict $I \cap N(S) = \varnothing$. Thus

$$|N(I)| \leq |N(S) \setminus S| = |N(S)| - |N(S) \cap S| < |S| - |N(S) \cap S| = |I|.$$

So an independent Hall obstruction exists whenever any Hall obstruction exists. This proves sufficiency. □

In particular, every negative certificate supplied for $n < 173$ in the associated JSON is already an independent vertex set: none of its members are adjacent to each other or carry a permitted loop. The criterion is global and does not depend on any conjectural cutoff.

Theorem 2 (Uniform near-perfect partial matchings)**.** *For every integer $n \geq 7$, write*

$$k = \lfloor (n+1)^{1/3} \rfloor, \qquad g(k) = (k+1)^3 - k^3 = 3k^2 + 3k + 1.$$

There is an explicitly constructible cube-sum partial permutation *matching at least*

$$\boxed{n - \left\lfloor \frac{g(k)}{2} \right\rfloor} \tag{3}$$

indices to distinct values within $[1, n]$. Consequently the maximum matching of the cube-sum bipartite graph has deficiency $O(n^{2/3})$ for every *n, including lengths where no complete cube-sum permutation exists.*

Proof. Choose whichever of the consecutive cubes $c_- = k^3$ and $c_+ = (k+1)^3$ is closest to $n+1$, and denote it by c. Their gap is $g(k)$, hence $\delta = |c - (n+1)| \leq \lfloor g(k)/2 \rfloor$. Since $n \geq 7$, $c_- \geq 8$. If $c_+ > 2n$, then $c_+ - (n+1) \geq n$ while $(n+1) - c_- \leq n - 7$, so the nearer cube must be c_-. Thus the chosen c lies in $[2, 2n]$.

The interval

$$J_c = [\max(1, c-n),\ \min(n, c-1)] \cap \mathbb{Z}$$

has precisely $n - \delta$ elements. It is invariant under $i \mapsto c - i$. On this interval set $p(i) = c - i$, giving a bijection $J_c \to J_c$ with the constant cube sum $i + p(i) = c$. Therefore it defines a partial perfect matching of size $n - \delta$. The claimed bound follows. □

Theorem 3 (Localization of Hall obstructions)**.** *For $n \geq 7$ let*

$$K = \lfloor (2n)^{1/3} \rfloor, \qquad G = 3K^2 + 3K + 1.$$

Every nonempty subset $S \subseteq [1, n]$ with diameter $\operatorname{diam}(S) = \max S - \min S$ *satisfying*

$$\boxed{\operatorname{diam}(S) \leq n - G}$$

automatically obeys Hall's condition $|N(S)| \geq |S|$ in the symmetric cube-sum bipartite graph. Consequently every *independent Hall obstruction must satisfy*

$$\boxed{\operatorname{diam}(S) > n - G,}$$

and must contain vertices from both the lower and upper boundary regions of width at most G.

Proof. Write $a = \min S$, $b = \max S$. The candidate integer interval

$$[b+1, a+n]$$

has $n - (b-a) \geq G$ consecutive integers, all between 2 and $2n$. Within this range there cannot be G consecutive noncubes: the difference between consecutive cubes is $3j^2 + 3j + 1 \leq G$ for $j \leq K$, while the initial integers $2, \ldots, 7$ also give fewer than G noncubes. Consequently some perfect cube c lies in $[b+1, a+n]$.

For every $i \in S$ we then have $1 \leq c - i \leq n$, and all $c - i$ are distinct. Thus $i \mapsto c - i$ is an injection of S into its neighborhood $N(S)$, proving Hall's condition. The contrapositive gives the claimed diameter restriction. The latter implies $\min S < G$ and $\max S > n - G + 1$, so a Hall obstruction has to meet both boundary regions. □

These theorems supply a completely noncomputational approach to the existence question. Instead of checking all subsets, one need only understand *long-spanning independent sets* whose vertices reach both ends of $[1, n]$. Eliminating those obstructions by a uniform argument would remove the remaining finite matching certificates; the near-perfect bound alone does not yet yield such an elimination.

Theorem 4 (Arithmetic codegrees and expansion)**.** *Let H_n be the cube-sum bipartite graph with left and right vertices $[1, n]$. For distinct left vertices $i \neq j$, their number of common right neighbors satisfies*

$$\boxed{|N(i) \cap N(j)| \leq \tau(|i - j|),} \tag{4}$$

where $\tau(d)$ is the number of positive divisors of d. Write

$$\delta_n = \min_i |N(i)|, \quad \Delta_n = \max_i |N(i)|, \quad D_n = \max_{1 \leq d < n} \tau(d).$$

For every nonempty $S \subseteq [1,n]$, with $s = |S|$,

$$\boxed{|N(S)| \geq \frac{s\,\delta_n^2}{\Delta_n + (s-1)D_n}.} \tag{5}$$

In particular, for each fixed $\varepsilon > 0$ there exist constants $c_\varepsilon > 0$ and N_ε such that any Hall-deficient set S in H_n, for $n \geq N_\varepsilon$, must satisfy

$$\boxed{|S| \geq c_\varepsilon n^{2/3-\varepsilon}.} \tag{6}$$

Proof. If v is adjacent to distinct left vertices $i > j$, there are cube roots $a > b$ with $i + v = a^3$, $j + v = b^3$. Therefore

$$i - j = a^3 - b^3 = (a-b)(a^2 + ab + b^2).$$

The positive integer $h = a - b$ must divide $i - j$. For a fixed $h > 0$, the function $b \mapsto (b+h)^3 - b^3$ is strictly increasing, hence at most one b solves the equation. This proves (4).

For a set S of s left vertices, put $r_v = |\{i \in S : i \sim v\}|$. Then

$$\sum_v r_v = \sum_{i \in S} |N(i)| \geq s\delta_n,$$

whereas the codegree estimate gives

$$\sum_v r_v^2 = \sum_{i \in S} |N(i)| + 2\sum_{\substack{i<j \\ i,j \in S}} |N(i) \cap N(j)| \leq s\Delta_n + s(s-1)D_n.$$

Cauchy–Schwarz on the nonzero r_v now yields (5).

For completeness, because the possible right neighbors of i correspond precisely to cubes in $[i+1, i+n]$,

$$|N(i)| = \lfloor (i+n)^{1/3} \rfloor - \lfloor i^{1/3} \rfloor.$$

Concavity of $x^{1/3}$ implies

$$\delta_n \geq (2^{1/3} - 1)n^{1/3} - 1, \qquad \Delta_n \leq \lfloor (2n)^{1/3} \rfloor.$$

The classical elementary divisor bound $\tau(d) = O_\varepsilon(d^\varepsilon)$ gives $D_n = O_\varepsilon(n^\varepsilon)$ for every fixed $\varepsilon > 0$. If S violates Hall, so $|N(S)| < s$, formula (5) forces

$$s > 1 + \frac{\delta_n^2 - \Delta_n}{D_n}.$$

Since δ_n^2 grows like $n^{2/3}$ and $\Delta_n = O(n^{1/3})$, this implies (6). □

Combining Theorem 4 with independent-set reduction and Theorem 3 gives a two-dimensional localization of any possible Hall obstruction: it must *both* contain $n^{2/3-o(1)}$ vertices and span a range of length $n - O(n^{2/3})$. These constraints hold whether or not an all-length existence threshold is already known. They do not, by themselves, rule out the remaining large independent sets.

Theorem 5 (Undirected cycle-cover expansion)**.** *If $\mathfrak{F}_n$ denotes the set of spanning cycle-cover configurations of G_n, and $c(F)$ is the number of its cycles of length at least three, then*

$$\boxed{P(n) = \sum_{F \in \mathfrak{F}_n} 2^{c(F)}.} \tag{7}$$

Thus $P(n) > 0$ if and only if G_n has a spanning union of loops, dimers, and simple cycles. In particular,

$$\boxed{P(n) \equiv I(n) \pmod 2,}$$

where $I(n)$ counts valid involutions.

Proof. Every valid permutation decomposes uniquely into fixed points, transpositions, and directed cycles of length at least three. The first correspond to permitted loops, the second to undirected dimer edges, and each longer undirected cycle admits exactly two orientations. Choices on different components are independent, giving (7). All terms with $c(F) > 0$ are even, whereas configurations with $c(F) = 0$ are exactly the involutions. □

Let H_n be the symmetric n-by-n zero-one matrix with $(H_n)_{ij} = 1$ precisely when $i + j$ is a perfect cube.

Corollary 1 (Efficient parity via binary algebra)**.**

$$\boxed{P(n) = \mathrm{per}(H_n), \qquad P(n) \bmod 2 = \det(H_n) \bmod 2.}$$

In particular $P(n)$ is odd exactly when H_n is nonsingular over $\mathbb{F}_2$, and its parity can be computed by polynomial-time Gaussian elimination without enumerating any permutations. Moreover,

$$\det(H_n) = 1 \pmod 2 \quad \Longrightarrow \quad \textit{there exists a legal involution}$$

since the number of such involutions is then odd. The converse need not hold.

Proof. The permanent expansion enumerates the admissible permutations. In characteristic two every permutation sign in the determinant is $+1$, so determinant and permanent coincide. □

Even more structure appears once any single legal permutation has been found. For such a permutation p, define the directed *alternation graph* D_p with vertex set $[1, n]$ and a non-loop arc $i \to j$ when $i + p(j)$ is a cube. Let $\mathcal{S}(D_p)$ be its strongly connected components, and let B_C be the square zero-one matrix indexed by $i, j \in C$, with $(B_C)_{ij} = 1$ if $i + p(j)$ is a cube (including all diagonal entries).

Theorem 6 (Exact strong-component factorization)**.** *For* every *legal reference permutation p of length n,*

$$\boxed{P(n) = \prod_{C \in \mathcal{S}(D_p)} \mathrm{per}(B_C).} \tag{8}$$

In particular $P(n) = 1$ precisely when every strong component is a singleton; and every nontrivial component contributes at least two possible permutations.

Proof. Any second legal permutation q determines a permutation $f = p^{-1} \circ q$ of $[1, n]$, and its arcs $i \to f(i)$ are allowed by the extended alternation graph with the identity loops included. Every cycle of f lies entirely inside one strongly connected component of D_p. Hence f restricts to independent permutations within the components, and their numbers multiply. Conversely, any such collection of internal permutation choices assembles into a legal q. The factorization follows. A nontrivial strong component contains a directed cycle, which can be toggled against the identity on that component, proving the final assertion. □

This theorem explains the previously computed $P(236) = 512$ from nine independent directed five-cycles and 191 singleton components. It also unifies the alternating-cycle criterion of Lemma 14 with exact counting. The companion verifier `global_graph_audit.py` checks the parity identity and component factorization in concrete, bounded-size instances using exact binary arithmetic.

3 Threshold-free renormalization and counting entropy

The specific all-n threshold is not needed for a powerful global closure property of the cube-sum permutation family.

Lemma 4 (Independent-fiber renormalization)**.** *Fix an integer exponent $d \geq 2$ and an odd positive integer t; set $L = t^d$, $A = (L - 1)/2$, and $T_t(m) = Lm + A$. Let $P_d(m)$ denote the number of permutations of $[1, m]$ whose index-image sums are positive perfect dth powers. Then for every nonnegative integer m,*

$$\boxed{P_d(T_t(m)) \geq P_d(A) P_d(m)^L,} \tag{9}$$

with $P_d(0) = 1$. Moreover,

$$2T_t(m) + 1 = L(2m + 1), \qquad T_s(T_t(m)) = T_{st}(m).$$

Proof. Partition $[1, T_t(m)]$ into a prefix $[1, A]$ and the remaining m consecutive blocks of length L. Choose an arbitrary valid permutation on the prefix, and choose independently L valid permutations $p_1, \dots, p_L$ of $[1, m]$. At $x = A + (i-1)L + r$, $1 \le r \le L$, define

$$q(x) = A + (p_r(i) - 1)L + (L + 1 - r).$$

This permutes the remaining interval and satisfies

$$x + q(x) = L(i + p_r(i)).$$

Each right-hand side is a perfect dth power, since $L = t^d$. Different prefix permutations and different tuples of p_r produce different full permutations, yielding (9). The identities for T_t follow directly from its affine definition. □

Lemma 5 (A four-interval 171-permutation)**.** *The following involution is a cube-sum permutation of* $[1, 171]$*:*

$$p(i) = \begin{cases} 8 - i, & 1 \le i \le 7, \\ 27 - i, & 8 \le i \le 19, \\ 64 - i, & 20 \le i \le 44, \\ 216 - i, & 45 \le i \le 171. \end{cases}$$

Proof. The four displayed intervals are pairwise disjoint and partition $[1, 171]$. Each is invariant under its indicated reflection, whose index-image sum is, respectively, $2^3, 3^3, 4^3$, or 6^3. □

Theorem 7 (Cubic affine closure and entropy monotonicity)**.** *For every integer* $m \ge 1$*,*

$$\boxed{P(343m + 171) \ge P(m)^{343}.}$$

Set $T(m) = 343m + 171$*. Then for each positive integer* k*,*

$$T^k(m) = 343^k\Big(m + \frac{1}{2}\Big) - \frac{1}{2}, \qquad \boxed{P(T^k(m)) \ge P(m)^{343^k}.}$$

For $P(m) > 0$*, define the normalized entropy*

$$h(m) = \frac{\log P(m)}{m + \frac{1}{2}}.$$

It is nondecreasing on each orbit:

$$\boxed{h(T^k(m)) \ge h(m).}$$

Consequently, without requiring any eventual-existence theorem,

$$\boxed{\limsup_{n \to \infty} \frac{\log P(n)}{n} \ge \sup_{\substack{m \ge 1 \\ P(m) > 0}} \frac{\log P(m)}{m + \frac{1}{2}}.} \tag{10}$$

Proof. Apply Lemma 4 with $d = 3$, $t = 7$, $L = 343$, and $A = (343 - 1)/2 = 171$. Lemma 5 gives $P(171) \ge 1$. Iterating the inequality and the affine composition law yields the second display. Since $T^k(m) + 1/2 = 343^k(m + 1/2)$, the entropy inequality follows on taking logarithms. For each fixed m with $P(m) > 0$, the orbit $T^k(m)$ tends to infinity, so its entropy gives a lower bound for the global limsup. Taking the supremum over all such m proves (10), interpreted in the extended-real sense if necessary. □

The transformation $2m + 1 \mapsto 343(2m + 1)$ is a multiplicative renormalization of the odd coordinate $2m + 1$. In particular the set of admissible lengths is closed under $m \mapsto 343m + 171$, independently of the numerical threshold in Theorem 18. The explicit arithmetic verifier `renormalization.py` checks both the four-block seed and independent fiber choices without relying on the large finite-certificate archive.

4 A single-seed eventual existence theorem

There is a substantially more conceptual route to *eventual existence* than proving the sharp cutoff at 173. It uses only one seven-element permutation, seven rounds of integer reflection closure, and a uniform cubic gap inequality. In particular, this proof never invokes a bipartite matching algorithm or any of the earlier 173–364 permutation certificates.

Let $\mathcal{R}_0 = \{7\}$ and define a sequence of finite sets recursively by

$$\mathcal{R}_{j+1} = \mathcal{R}_j \cup \{k^3 - m - 1 : \ m \in \mathcal{R}_j, \ m < k^3 - m - 1 \le 10975, \ k \ge 2\}. \tag{11}$$

Lemma 6 (Seven-round arithmetic certificate)**.** *For $j = 0, \dots, 7$ the exact cardinalities $|\mathcal{R}_j|$ are*

$$1, \ 21, \ 189, \ 995, \ 3280, \ 6588, \ 8533, \ 8924.$$

Moreover,

$$\boxed{[8915, 10975] \cap \mathbb{Z} \subseteq \mathcal{R}_7.}$$

Proof. This is a bounded integer-only calculation, reproduced by `one_seed_eventual.py`. For each member m and each integer $k \ge 2$ with $k^3 \le 21950$, form $n = k^3 - m - 1$ and insert it when $m < n \le 10975$. After seven simultaneous rounds the cardinalities are exactly those displayed and all 2061 required lengths are present. The computation tests only elementary integer identities and finite set membership, not the existence of matchings. Independently, the verifier `one_seed_interval_proof.py` operates entirely on unions of integer intervals. For every current interval $[a, b]$ and cube c, it restricts to

$$\ell = \max(a, c - 10976), \qquad h = \min(b, \lfloor (c-2)/2 \rfloor)$$

and, if $\ell \le h$, adds the reflected interval $[c - h - 1, c - \ell - 1]$. The exact numbers of disjoint intervals after rounds 0 through 7 are

$$1, \ 21, \ 186, \ 870, \ 2077, \ 2037, \ 1031, \ 770.$$

The last interval of the final union is exactly $[8915, 10975]$, which proves the finite coverage using *interval arithmetic alone*. Both verifiers reproduce the same seven-stage cardinalities. This finite check is separate from, and much weaker than, a search for perfect matchings. □

Theorem 8 (Eventual existence from one permutation)**.** *The explicit permutation*

$$\boxed{p = (7, 6, 5, 4, 3, 2, 1)}$$

at length 7, whose cube sums all equal 8, together with the reflection-extension lemma, proves that

$$\boxed{P(n) > 0 \quad \textit{for every } n \ge 8915.}$$

Thus eventual existence requires only one elementary starting permutation and the seven-round arithmetic certificate, without locating the optimal cutoff.

Proof. Each element of $\mathcal{R}_j$ is a length possessing a valid cube-sum permutation, by induction on j using Lemma 1. Therefore the finite interval $8915 \le n \le 10975$ is covered.

For $n \ge 10976$, let k be the least integer such that $k^3 \ge n + 8916$ and put $m = k^3 - n - 1$. Then $m \ge 8915$, and $k \ge 28$. We prove $k^3 \le 2n$. If $k = 28$, this follows from $n \ge 10976$, since $28^3 = 21952$. If $k \ge 29$, minimality gives $n \ge (k-1)^3 - 8915$, whence

$$\begin{aligned} 2n - k^3 &\ge 2((k-1)^3 - 8915) - k^3 \\ &= k^3 - 6k^2 + 6k - 17832 \\ &\ge 1685 > 0. \end{aligned}$$

The last polynomial equals 1685 at $k = 29$ and is strictly increasing for all larger integer k. Thus $8915 \le m < n$. By strong induction $P(m) > 0$, so reflection through the cube k^3 produces a valid permutation of length n. This proves the theorem. □

Theorem 9 (Eventual existence of involutions)**.** *Let $I_4(n)$ denote the number of cube-sum* involutions *of $[1, n]$ satisfying $p(4) = 4$, where an involution obeys $p^2 = \mathrm{id}$. Then*

$$\boxed{I_4(n) \geq F(n) > 0 \qquad \text{for every } n \geq 8915.}$$

More strongly,

$$\boxed{\liminf_{n\to\infty} \frac{\log I_4(n)}{(2n)^{1/3}} \geq \frac{\log 2}{2}.} \tag{12}$$

In particular, the cube-sum graph admits a spanning configuration consisting entirely of loops and disjoint dimers at every $n \geq 8915$.

Proof. The sole starting permutation $p(i) = 8 - i$ on $[1, 7]$ is an involution and fixes the index 4. Every suffix reflection $i \mapsto c - i$ in Lemma 1 is itself an involution of the new interval, which is disjoint from all previously defined positions. Thus every constructed permutation in the one-seed reflection tree is an involution fixing 4. Theorem 8 proves $F(n) > 0$ for $n \geq 8915$, and the uniqueness of reflection histories proved in Lemma 8 gives $I_4(n) \geq F(n)$. Finally, Theorem 12 supplies (12). The undirected loop-and-dimer formulation follows by decomposing an involution into fixed points and transpositions. □

This separates two logically different facts. The 236-element cube-sum graph admits 512 ordinary permutations but no involutions, by Theorem 20. Nevertheless, the one-seed method proves that noninvolutive obstructions cannot persist for all large lengths. It does so without any matching solver or sharp-cutoff data.

5 Parity-optimal fixed-point structure

The single-seed eventual-involution theorem can be strengthened from the existence of an involution to the existence of an involution with the *minimum possible number of fixed points*. This does not follow from the 173-threshold theorem or from general perfect 2-matchings: it uses parity-preserving odd-cube reflections and two tiny explicit seeds.

A reflection step from m to n through an *odd* cube $c = k^3$ satisfies

$$n = c - m - 1 \equiv m \pmod 2.$$

Its new interval reflection $i \mapsto c - i$ has no fixed point, since $c/2 \notin \mathbb{Z}$. Thus the parity of the length and the exact set of fixed points are preserved under every odd-cube extension.

Lemma 7 (Two parity-specific finite closures)**.** *Starting from the involution $p_7(i) = 8 - i$ on $[1, 7]$ and allowing strict extensions through* odd *cubes only, the reachable lengths include* every odd integer *in*

$$\boxed{[70001, 100001].}$$

Starting from the fixed-point-free involution $p_{26}(i) = 27 - i$ on $[1, 26]$, the same odd-cube reflection rule reaches every even integer *in*

$$\boxed{[230000, 302000].}$$

Proof. This is an exact finite integer calculation, reproduced by `parity_minimal_involutions.py`. For each seed a and even or odd candidate n of matching parity, iterate the rule

$$R_a(a) = 1, \quad R_a(n) = \bigvee_{\substack{k\geq 3 \text{ odd} \\ a \leq k^3 - n - 1 < n}} R_a(k^3 - n - 1).$$

The two consecutive-parity intervals contain respectively 15001 and 36001 certified lengths. The last smaller parity-compatible lengths which are *not* reachable below the respective interval endpoints are 69601 (odd) and 228026 (even). A second, algorithmically independent checker `parity_interval_proof.py` works with *unions of integer intervals* in the coordinate $t = (n - p)/2$ where $p \in \{0, 1\}$ is the length parity. An odd-cube reflection becomes $t \mapsto d - t$ with $d = (k^3 - 1 - 2p)/2$. For every current interval $[a, b]$, the strict-extension constraints give the exactly reflected interval

$$\left[d - \min\left(b, \left\lfloor \tfrac{d-1}{2} \right\rfloor\right), d - \max(a, d - T_{\max})\right]$$

when the lower and upper source bounds are consistent, with $T_{\max} = (H - p)/2$. After just twelve simultaneous rounds, the merged unions contain both claimed parity-specific coverage intervals. The resulting numbers of reachable parity lengths are respectively 41057 and 121707, in 2997 and 7047 disjoint integer intervals. The first checker also reconstructs full involutions at the endpoints and verifies the exact fixed point sets. Neither checker uses a matching solver or the sharp-threshold certificates. □

Theorem 10 (Eventual parity-minimal involutions)**.** *For every odd integer* $n \geq 70001$ *there exists a legal cube-sum involution on* $[1, n]$ *whose* unique *fixed point is* 4*. For every even integer* $n \geq 230000$ *there exists a legal cube-sum involution of* $[1, n]$ *with* no fixed point*. Equivalently,*

$$\boxed{\begin{array}{ll} n \geq 70001 \text{ odd}: & \min_{p^2=\mathrm{id}} \#\mathrm{Fix}(p) = 1, \\ n \geq 230000 \text{ even}: & \min_{p^2=\mathrm{id}} \#\mathrm{Fix}(p) = 0, \end{array}}$$

where the minima are over legal cube-sum involutions.

Proof. The preceding lemma supplies the finite intervals and the corresponding minimal-fixed-point involutions, because odd-cube reflections preserve the set of fixed points.

For either parity put $(M, H) = (70001, 100001)$ or $(230000, 302000)$, respectively. If $n > H$ has the required parity, take the *least odd* k with

$$k^3 \geq n + M + 1, \qquad m = k^3 - n - 1.$$

Then $m \geq M$, and m has the same parity as n. We show $m < n$. Since k is the least such odd root, $n \geq (k-2)^3 - M$. Therefore

$$2n - k^3 \geq 2(k-2)^3 - k^3 - 2M = k^3 - 12k^2 + 24k - 16 - 2M.$$

For the odd case $n \geq 100003$, hence $k \geq 57$, and the final polynomial is increasing for $k \geq 57$ and is $7555 > 0$ at $k = 57$. For the even case $n \geq 302002$, we have $k \geq 83$; its value at $k = 83$ is $31095 > 0$ and the polynomial is again increasing. Consequently $k^3 \leq 2n$, and $M \leq m < n$.

Strong induction within the fixed parity now produces a minimal-fixed-point involution at m, followed by the odd-cube suffix reflection from m to n. This adds no fixed points. The parity lower bounds (one fixed point for odd n, none for even n) are tautological for involutions, so the achieved counts are minimal. □

Corollary 2 (Odd-cube perfect matchings for every large even length)**.** *For every even* $n \geq 230000$*, there is a fixed-point-free involution* p *such that* every $i + p(i)$ *is an* odd *perfect cube. Equivalently, the odd and even integers of* $[1, n]$ *can be paired bijectively using edges whose endpoint sums are odd cubes.*

For every odd $n \geq 70001$ *there is a cube-sum involution with exactly one fixed point at* 4*, whose cube sums are odd at* all *indices* $i \geq 8$*; the first seven sums are all* 8*.*

Proof. The even seed $p_{26}(i) = 27 - i$ has only the odd cube sum 27, and every subsequent extension uses an odd cube. The final involution is fixed-point-free, so each transposition joins an odd integer to an even integer.

The odd seed $p_7(i) = 8 - i$ has precisely seven positions with the even cube sum 8, and later odd-cube reflections introduce no further even cube sums. Theorem 10 supplies the required constructions at all sufficiently large lengths. □

Define $J_{\min}(n)$ to count legal cube-sum involutions with no fixed point when n is even, or with exactly one fixed point at 4 when n is odd.

Theorem 11 (Uniform entropy of minimally fixed involutions)**.** *For all sufficiently large* n*,*

$$\boxed{J_{\min}(n) \geq \exp\bigl(\Omega(n^{1/3})\bigr).}$$

More precisely,

$$\boxed{\liminf_{n\to\infty} \frac{\log J_{\min}(n)}{(2n)^{1/3}} \geq \frac{\log 2}{4}.} \tag{13}$$

Proof. For $a \in \{7, 26\}$, let $F_a^{\mathrm{odd}}(n)$ count the distinct strict-reflection histories from seed a which use *odd* cube roots only. All such histories have the seed's parity and fixed-point structure; their permutations are distinct by Lemma 8. Theorem 10 implies $F_a^{\mathrm{odd}}(n) \geq 1$ on its corresponding parity tail $n \geq M_a$.

Fix $s \geq 2$, let ρ_s solve $\sum_{h=1}^{s} \rho_s^{-h} = 1$, and set

$$B_s = \max\{8(s-1)^2 + 24(s-1) + 15,\ 4s + 80\}.$$

For $n \geq M_a$, put $k = \lfloor (2n)^{1/3} \rfloor$. We prove by strong induction

$$F_a^{\mathrm{odd}}(n) \geq \rho_s^{(k-B_s)/4}.$$

This is immediate if $k < B_s$ by positivity. Otherwise, let q be the largest odd integer with $q \leq k$ and use the s distinct odd cubes

$$c_j = (q-2j)^3, \qquad 0 \leq j < s.$$

Their predecessors $m_j = c_j - n - 1$ have the same parity as n. Since $2n < (k+1)^3$ and $q \geq k-1$,

$$\begin{aligned} 2m_j - (k-4j-4)^3 &\geq 2(k-2j-1)^3 - (k+1)^3 - 1 - (k-4j-4)^3 \\ &= 3k^2 - (24j^2 + 72j + 45)k \\ &\quad + 48j^3 + 168j^2 + 180j + 60 \geq 0. \end{aligned}$$

The last inequality holds by $k \geq B_s \geq 8j^2 + 24j + 15$. As $k - 4s \geq 80$, it follows that $m_j \geq 80^3/2 = 256000 \geq M_a$. Also $c_j \leq k^3 \leq 2n$, so $m_j < n$. Writing $K(m) = \lfloor (2m)^{1/3} \rfloor$, we have $K(m_j) \geq k - 4(j+1)$. Consequently, strong induction and the exact reflection counting recurrence give

$$\begin{aligned} F_a^{\mathrm{odd}}(n) &\geq \sum_{j=0}^{s-1} F_a^{\mathrm{odd}}(m_j) \\ &\geq \rho_s^{(k-B_s)/4} \sum_{j=0}^{s-1} \rho_s^{-(j+1)} = \rho_s^{(k-B_s)/4}. \end{aligned}$$

For fixed s, the constant B_s is independent of n. Letting $n \to \infty$ in each parity, followed by $s \to \infty$ and $\rho_s \uparrow 2$, proves (13), because $J_{\min}(n)$ dominates the corresponding $F_a^{\mathrm{odd}}(n)$. □

This shows that the minimal fixed-point restriction costs at most a change in the proved cube-root entropy coefficient: it does not destroy uniform superpolynomial combinatorial multiplicity. The direct taxicab family in Section 15 separately establishes stronger *linear* entropy on an infinite subsequence.

Corollary 3 (Efficient one-seed construction)**.** *For every $n \geq 8915$, a cube-sum permutation of length n is constructible using $O(n)$ unit-cost arithmetic operations and $O(\log \log n)$ reflection layers beyond a fixed finite lookup.*

Proof. For $n \geq 10976$, take the least cube $k^3 \geq n + 8916$ as in the theorem. Its predecessor satisfies

$$8915 \leq m = k^3 - n - 1 = O(n^{2/3}),$$

since consecutive cubes differ by $O(k^2) = O(n^{2/3})$. Hence $\log m \leq (2/3) \log n + O(1)$, which yields $O(\log \log n)$ recursive stages before reaching the finite interval. Every reflection writes its newly added positions exactly once, so the output work is $O(n)$; exact cube roots add only polylogarithmic overhead. The finite interval is traced through at most seven predecessor links, ending at the sole seed of length 7. □

Theorem 8 is weaker numerically than the fully classified cutoff in Theorem 18, but stronger conceptually in a different direction: it establishes eventual existence without any large explicit matching witness. The residual finite verification concerns *reflection reachability only*, not bipartite matching, and is entirely independent of the sharp-threshold classification. A completely computation-free proof of eventual existence would still require eliminating this last seven-round finite arithmetic check.

6 Monotone signed Waring representations

The reflection orbit has a direct reformulation as an additive-number-theoretic problem. The important feature is not just the presence of signed cubes, but the *monotonicity of every intermediate partial reflection.*

Proposition 1 (Reflection paths and alternating cubes)**.** *Fix any positive starting length a and set $x_0 = 2a+1$. A sequence of r strict reflection extensions from a to n exists if and only if there are integers $k_1, \dots, k_r \geq 2$ for which*

$$x_j = 2k_j^3 - x_{j-1} > x_{j-1} \qquad (1 \leq j \leq r),$$

and $x_r = 2n+1$. Equivalently,

$$\boxed{2n+1 = (-1)^r(2a+1) + 2\sum_{j=1}^{r}(-1)^{r-j}k_j^3,}$$

where the intermediate quantities specified by the recursion must be strictly increasing positive odd integers.

Proof. Writing $x_j = 2n_j + 1$, the reflection condition $n_{j-1} + n_j + 1 = k_j^3$ is equivalent to $x_{j-1} + x_j = 2k_j^3$. The strict extension $n_j > n_{j-1}$ is exactly $x_j > x_{j-1}$. Iterating this affine identity gives the alternating sum formula. Conversely every such increasing sequence recovers the necessary reflection predecessors and cubes. □

For the canonical seed $a = 7$, a path of exactly seven reflections reaches n precisely when it admits

$$\boxed{n + 8 = k_7^3 - k_6^3 + k_5^3 - k_4^3 + k_3^3 - k_2^3 + k_1^3}$$

with the corresponding strictly increasing intermediate odd coordinates. This converts the global question about *bounded reflection depth* into an explicit restricted signed-Waring representation problem. It is stronger than mere eventual existence: Theorem 8 proves a growing but $O(\log\log n)$-bounded depth, not a universal depth of seven.

Experiments with exact interval arithmetic suggest that the seven-step orbit already covers every length $n \geq 8915$, but this stronger unbounded statement is *not proved here.* It would require a uniform theorem about alternating cubic representations with their intermediate-order constraints, rather than an unbounded numerical extrapolation.

7 The reflection tree as a standalone counting model

The one-seed construction contains not merely one permutation for every sufficiently large length, but a recursively countable subfamily whose size itself grows uniformly.

Let $F(7) = 1$, and for $n > 7$ define

$$\boxed{F(n) = \sum_{\substack{7 \leq m < n \\ m+n+1 = k^3 \text{ for some } k \geq 2}} F(m).} \tag{14}$$

We regard $F(n)$ as zero at lengths not reachable from the seven-element seed.

Lemma 8 (Exact reflection-history enumeration)**.** *For every $n \geq 7$, $F(n)$ equals exactly the number of distinct cube-sum permutations obtainable from the seed $(7,6,5,4,3,2,1)$ by a finite sequence of strict reflection extensions. In particular,*

$$\boxed{0 \leq F(n) \leq P(n), \qquad F(n) > 0 \quad (n \geq 8915).}$$

Proof. Every strictly increasing history has a unique final predecessor $m < n$, and its last extension uses the cube $c = m + n + 1$. The final entry of the permutation is then

$$p_n = c - n = m + 1.$$

Thus the last predecessor m can be recovered uniquely from p_n, while the previous permutation is recovered by restricting to its first m positions. Hence two distinct reflection histories cannot produce the same final permutation. The recursive count is exactly (14). The eventual positivity is Theorem 8. □

Theorem 12 (Uniform entropy inside one reflection tree)**.** *Fix $s \geq 2$ and let $\rho_s \in (1,2)$ be determined by $\sum_{h=1}^{s} \rho_s^{-h} = 1$. Put*

$$B_s^* = \max\{2s^2 + 4s - 1,\ 2s + 27\}, \qquad K(n) = \lfloor \sqrt[3]{2n} \rfloor.$$

For every $n \geq 8915$,

$$\boxed{F(n) \geq \rho_s^{(K(n) - B_s^*)/2}.}$$

Consequently,

$$\boxed{\liminf_{n\to\infty} \frac{\log F(n)}{(2n)^{1/3}} \geq \frac{\log 2}{2}.} \tag{15}$$

The same bound holds for $P(n)$, but this proof uses only the seven-element seed, its seven-round arithmetic reachability certificate, and elementary cubic inequalities.

Proof. Write $k = K(n)$, $\rho = \rho_s$, and $B = B_s^*$. If $k < B$, the right-hand side of the claimed inequality is at most 1, and $F(n) \geq 1$ by Lemma 8.

If $k \geq B$, for $j = 0, \ldots, s-1$ take $c_j = (k-j)^3$ and $m_j = c_j - n - 1$. The polynomial inequality in the proof of Theorem 29 (applied with the same j, k) gives

$$K(m_j) \geq k - 2j - 2.$$

Since $k - 2s \geq 27$, this guarantees $m_j \geq \lceil 27^3/2 \rceil > 8915$. Also $c_j \leq k^3 \leq 2n$, so $m_j < n$. By strong induction, the exact recurrence (14) therefore yields

$$\begin{aligned} F(n) &\geq \sum_{j=0}^{s-1} F(m_j) \\ &\geq \rho^{(k-B)/2} \sum_{j=0}^{s-1} \rho^{-(j+1)} = \rho^{(k-B)/2}. \end{aligned}$$

Finally, for every fixed s, B_s^* does not depend on n, and $\rho_s \uparrow 2$ as $s \to \infty$. The limit inferior is at least $(\log \rho_s)/2$ for each s, which proves (15). □

The reflection-tree recurrence is a useful independent combinatorial object: its members are explicit permutations, not merely abstract paths or potential matchings. Unlike a claim that every large permutation can be generated by reflection (which is false, for example at $n = 173$ from this seed), this theorem counts a rigorously defined *subfamily* only.

8 The entropy barrier of pure reflection

The reflection-tree estimates admit a sharp structural complement: its combinatorial entropy has exactly the cube-root growth *order*. Consequently, the exponentially many permutations in the independent-cycle constructions cannot be produced from finitely many seeds by suffix reflections alone.

Theorem 13 (Strict ordering of reflection cubes)**.** *Consider any chain of strict reflection extensions*

$$n_0 < n_1 < \cdots < n_r = n$$

whose step from n_{j-1} to n_j uses the cube $c_j = n_{j-1} + n_j + 1$. Then

$$\boxed{c_1 < c_2 < \cdots < c_r \leq 2n.}$$

If the initial seed permutation is fixed, the set $\{c_1, \ldots, c_r\}$ uniquely determines the entire history and its final permutation.

Proof. The step from n_{j-1} to n_j is strict, hence $c_j = n_{j-1} + n_j + 1 \leq 2n_j \leq 2n$. For consecutive steps,

$$c_{j+1} - c_j = n_{j+1} - n_{j-1} > 0.$$

Once the starting length n_0 and the strictly increasing list of cubes are specified, every subsequent length is uniquely determined by $n_j = c_j - n_{j-1} - 1$. The reflected permutation is then uniquely determined by the starting permutation and these successive lengths. A set of cubes therefore determines at most one history. □

Theorem 14 (Optimal cube-root order for the one-seed tree)**.** *Let $F(n)$ count cube-sum permutations obtained by strict reflection extensions from the unique seed $(7,6,5,4,3,2,1)$. For all $n \geq 7$,*

$$\boxed{F(n) \leq 2^{K(n)-1}, \qquad K(n) = \lfloor (2n)^{1/3} \rfloor.} \tag{16}$$

Together with Theorem 12, this yields

$$\boxed{\log F(n) = \Theta(n^{1/3}) \qquad (n \to \infty).}$$

The same upper estimate holds with a factor S for the union of strict-reflection descendants of any fixed finite collection of S initial cube-sum permutations:

$$\boxed{|\mathcal{R}_{\mathcal{S}}(n)| \leq S\, 2^{K(n)-1}.} \tag{17}$$

Proof. All possible strict reflection cubes up to length n are among $2^3, 3^3, \ldots, K(n)^3$, a list of $K(n)-1$ cubes. By Theorem 13, every reflection history from a fixed seed corresponds injectively to a subset of these cube roots (including the empty subset for the initial length). There are at most $2^{K(n)-1}$ such subsets. Summing over S seeds gives (17). Theorem 12 provides a strictly positive lower liminf for $\log F(n)/(2n)^{1/3}$ as $n \to \infty$, and the upper bound gives a finite limsup. Hence the logarithm is $\Theta(n^{1/3})$. □

Corollary 4 (A strict entropy separation)**.** *For any fixed finite family of cube-sum seed permutations,*

$$\lim_{n\to\infty} \frac{1}{n} \log\bigl(1 + |\mathcal{R}_{\mathcal{S}}(n)|\bigr) = 0.$$

By contrast, the entire legal family has positive upper exponential entropy:

$$\boxed{\limsup_{n\to\infty} \frac{\log P(n)}{n} \geq \frac{148877 \log 2}{681707} > 0.}$$

Consequently, no fixed finite-seed strict-reflection scheme can capture a positive fraction of the exponential-scale freedom exhibited by the independent three-cycle family.

Proof. The upper estimate (17) gives $\log(1 + |\mathcal{R}_{\mathcal{S}}(n)|) = O(n^{1/3}) = o(n)$. The lower bound for the full family is Theorem 22. In particular, on its explicit subsequence $n = 681707u^3$, the logarithm of the full permutation count is at least a positive constant times n, while the reflected finite-seed subfamily has logarithm $o(n)$. □

More generally, with perfect dth powers in place of cubes, the strictly increasing-root proof gives at most

$$S\, 2^{\lfloor (2n)^{1/d} \rfloor - 1}$$

permutations from S fixed seeds. The exponent $n^{1/d}$ is a *methodological upper bound* for reflection-only generation, not an upper bound for all admissible power-sum permutations.

Theorem 15 (Fibonacci majorant for the reflection tree)**.** *Let $F(n)$ be the number of strict-reflection descendants of the seven-point seed, as in* (14)*, and $K(n) = \lfloor (2n)^{1/3} \rfloor$. With $\mathrm{Fib}_1 = \mathrm{Fib}_2 = 1$ and $\mathrm{Fib}_{j+2} = \mathrm{Fib}_{j+1} + \mathrm{Fib}_j$,*

$$\boxed{F(n) \leq \mathrm{Fib}_{K(n)-2} \qquad (K(n) \geq 3),} \tag{18}$$

and $F(n) \leq 1$ when $K(n) = 2$. In particular, writing $\varphi = (1+\sqrt{5})/2$,

$$\boxed{\limsup_{n\to\infty} \frac{\log F(n)}{(2n)^{1/3}} \leq \log \varphi.} \tag{19}$$

Combined with Theorem 12,

$$\boxed{\frac{\log 2}{2} \leq \liminf_{n\to\infty} \frac{\log F(n)}{(2n)^{1/3}} \leq \limsup_{n\to\infty} \frac{\log F(n)}{(2n)^{1/3}} \leq \log \varphi.}$$

Proof. Write $k = K(n)$, and let $m = (k-j)^3 - n - 1 \geq 7$ be any valid strict predecessor appearing in (14). Necessarily $j \geq 0$ and $(k-j)^3 > n \geq k^3/2$, so $k - 2j > 0$. Moreover,

$$\begin{aligned} 2m &= 2(k-j)^3 - 2n - 2 \\ &\leq 2(k-j)^3 - k^3 - 2 \\ &= (k-2j)^3 - 6j^2(k-j) - 2 \\ &< (k-2j)^3. \end{aligned}$$

Thus the integer cube-root height of every predecessor satisfies

$$\boxed{K(m) \leq k - (2j+1).}$$

Define $B_\ell = 0$ for $\ell \leq 1$, $B_2 = 1$, and for $\ell \geq 3$ set

$$B_\ell = \sum_{j \geq 0} B_{\ell - 2j - 1}.$$

All sums here are finite. The recurrence gives $B_3 = B_4 = 1$ and $B_\ell = B_{\ell-1} + B_{\ell-2}$ for every $\ell \geq 5$; hence $B_\ell = \mathrm{Fib}_{\ell-2}$ for every $\ell \geq 3$. In particular B_ℓ is nondecreasing for $\ell \geq 2$.

Strong induction on n gives $F(n) \leq B_{K(n)}$: the seed $n = 7$ has height $K(7) = 2$ and $F(7) = B_2 = 1$; other lengths with height two have $F(n) = 0$; and for $K(n) = k \geq 3$,

$$F(n) \leq \sum_{j \geq 0} B_{K(m_j)} \leq \sum_{j \geq 0} B_{k-(2j+1)} = B_k.$$

This proves the finite inequality. Finally Binet's formula implies $\log \mathrm{Fib}_k = k \log \varphi + O(1)$, so (19) follows. The lower inequality is Theorem 12. □

This replaces the coarse $2^{K(n)-1}$ encoding ceiling with a strict Fibonacci bound. It captures the important fact that the distinct predecessor branches do not just consume one increasing cube root: they reduce the available height by the odd increments $1, 3, 5, \ldots$. It remains an open question whether the exact upper or lower limiting coefficients can be attained.

9 A dimension barrier for bounded reflection depth

Even though the seven-element seed reaches every sufficiently large length, a genuinely *uniformly bounded* number of reflections is a separate Diophantine question. There is an elementary counting obstruction to any bound of at most three reflections.

Theorem 16 (Reflection-depth sparsity)**.** *Fix one cube-sum permutation as an initial seed, and let $\mathcal{D}_r(N)$ be the set of output lengths at most N obtainable from it by at most r strict reflection extensions. For $K = \lfloor (2N)^{1/3} \rfloor$,*

$$\boxed{|\mathcal{D}_r(N)| \leq \sum_{j=0}^{r} \binom{K-1}{j}.} \tag{20}$$

For each fixed $r \leq 2$,

$$\boxed{|\mathcal{D}_r(N)| = O(N^{r/3}) = o(N),}$$

and for three or fewer reflections,

$$\boxed{|\mathcal{D}_3(N)| \leq \frac{N}{3} + O(N^{2/3}).} \tag{21}$$

In particular, no one-seed reflection construction of depth at most three can cover every sufficiently large length. At least four strict reflections are necessary for any one-seed, bounded-depth cofinite construction.

Proof. By Theorem 13, a history producing a length $n \le N$ uses a strictly increasing sequence of cubes chosen from the $K-1$ possible cubes $2^3, \dots, K^3$. A history with exactly j steps is thus encoded by a j-element subset, and there are at most $\binom{K-1}{j}$ choices. Summing yields (20). For fixed $r \le 2$ this is $O(K^r) = O(N^{r/3})$. For $r = 3$, the sum is

$$\binom{K-1}{3} + O(K^2) = \frac{K^3}{6} + O(K^2) \le \frac{N}{3} + O(N^{2/3}),$$

because $K^3 \le 2N$. Thus at least $(2/3+o(1))N$ of the first N lengths cannot be reached in at most three steps. □

More generally, if a fixed finite family of S initial permutations is allowed, the right side of (20) is multiplied by at most S. This makes clear why the counting barrier does not extend unchanged to arbitrarily many input seeds.

The numerical seven-round closure in Lemma 6 raises the sharper question whether *a fixed depth of seven* already suffices for all sufficiently large lengths. The present theorem does not prove this; it places a rigorous lower bound of four on any such single-seed uniform depth.

10 A hand-checkable finite interval proof

We give a standalone *finite algebraic* certificate for every base length $173 \le n \le 364$. It is independent of the augmenting-path search, Hall tests, and permutation matching algorithms. Unlike the global theory, this part still uses an explicit finite case table: the numerical expressions below can each be verified solely by integer arithmetic.

Lemma 9 (Boundary cancellation criterion)**.** *Let $n \ge 1$ and choose a positive integer t, endpoints $0 = b_0 < b_1 < \cdots < b_t = n$, and perfect cubes $c_1, \dots, c_t$. Define*

$$L_j = c_j - b_j, \qquad R_j = c_j - b_{j-1}.$$

Suppose that $1 \le L_j < R_j \le n+1$ for all j. Then the piecewise formula

$$\boxed{p(i) = c_j - i \quad \text{whenever } b_{j-1} < i \le b_j}$$

is a cube-sum permutation of $[1, n]$ if and only if

$$\boxed{\{1, R_1, \dots, R_t\}_{\text{multi}} = \{n+1, L_1, \dots, L_t\}_{\text{multi}}.} \tag{22}$$

Proof. The jth input segment is mapped bijectively onto the consecutive integers $[L_j, R_j - 1]$, with $i + p(i) = c_j$. Let $f(x)$ count the number of image segments containing the integer x, for $1 \le x \le n$. Set $f(0) = f(n+1) = 0$. The difference $f(x) - f(x-1)$ is the number of left boundaries at x minus the number of right boundaries at x, where the image intervals are half-open $[L_j, R_j)$. The multiset identity (22) is precisely the assertion that this difference is 1 at $x = 1$, -1 at $x = n+1$, and 0 at all other integers. Hence $f(x) = 1$ at every $1 \le x \le n$: the image intervals partition $[1, n]$. Conversely, a partition has precisely those boundary differences, and thus satisfies the stated multiset identity. □

Lemma 10 (Reflected-block rearrangement)**.** *Choose positive input-block lengths $\ell_1, \dots, \ell_t$ of sum n and an arbitrary output order π of these t blocks. Let A_j be the sum of input lengths before block j, and let B_j be the sum of output lengths before block j in order π. If all numbers*

$$C_j = 1 + A_j + B_j + \ell_j$$

are perfect cubes, then on input block j the reflection $p(i) = C_j - i$ gives a cube-sum permutation of $[1, n]$.

Proof. The input block is $[A_j + 1, A_j + \ell_j]$, which maps exactly to the prescribed output block $[B_j + 1, B_j + \ell_j]$. Those output blocks partition $[1, n]$ by construction. Each index sum is C_j. □

Family I (four blocks). Let $U < V$ be integer cubes and put

$$a = 2U - V - 1, \qquad b = 2V - 3U + 1.$$

If $a, b > 0$, use input lengths (a, b, a, b) and output order $(3, 4, 1, 2)$. Lemma 10 gives the cubic sums (U, V, U, V) and length $2(V - U)$. The pairs $(U, V) = (64, 125), (125, 216), (216, 343)$ yield lengths $122, 182, 254$ respectively.

Family II (five blocks). For cubes $X < Y < Z$, define

$$a = 2Y - Z - 1, \quad b = X + 2Z - 4Y + 1, \quad c = Y - X.$$

If $a, b, c > 0$, the lengths (a, b, c, a, c) in output order $(4, 2, 5, 1, 3)$ produce sums (Y, X, Z, Y, Z) and total length $2Y - X - 1$. With $(Y, Z) = (64, 125)$ and $X = 27, 8$, the results are $n = 100, 119$.

Family III (six blocks). For integer cubes X, W, Y, Z, put

$$\begin{aligned} a &= 2Y - Z - 1, & b &= X - 2a - 1, \\ c &= W + 2a - 2X + 1, & d &= Y + X - W - 2a - 1. \end{aligned}$$

If these are positive, choose lengths (a, b, c, d, a, d) and output order $(5, 2, 3, 6, 1, 4)$. The cubic sums are (Y, X, W, Z, Y, Z) and the length is $2Z - 2Y + X - W$. Taking $(Y, Z) = (64, 125)$, the choices $(X, W) = (8, 64), (27, 64), (8, 27)$ give $n = 66, 85, 103$, respectively.

Lemma 11 (Affine-word rigidity)**.** *Fix a finite list of integer cubes $C_1, \ldots, C_t$. Suppose all breakpoints $b_j(n)$ are affine functions of n with $b_0(n) = 0$ and $b_t(n) = n$. The resulting fixed-cube reflection word can be a valid permutation for at most two distinct values of n.*

Proof. Every legal permutation satisfies the sum identity

$$\sum_{j=1}^{t} C_j \big(b_j(n) - b_{j-1}(n)\big) = \sum_{i=1}^{n} (i + p(i)) = n(n+1).$$

Its left side is affine in n; its right side is a monic quadratic. The equality cannot hold for three distinct values. □

The obstruction concerns *fixed* cube sums and affine endpoints only, not varying cube sums or composition of reflection templates.

In total these *three symbolic identities* give eight base lengths

$$66, 85, 100, 103, 119, 122, 182, 254$$

without eight separate endpoint-table checks.

Theorem 17 (Reduced finite arithmetic certificate)**.** *The 74 integer identities in Appendix A, together with three symbolic block families, prove that every $n \geq 173$ admits a cube-sum permutation. The independent finite data consist of 74 endpoint identities (969 blocks); eight further base lengths are consequences of parameterized algebraic identities.*

Proof. The appendix contains 16 exceptional seed identities below 173 and 58 in the target range $173 \leq n \leq 364$. Lemma 9 makes each a legal permutation.

The three preceding formulas establish another six seeds below 173, namely $66, 85, 100, 103, 119, 122$, and two seeds in the target interval, $182, 254$. Their permutations are guaranteed by the symbolic block-rearrangement lemma. Thus 22 lower and 60 middle seeds have been established without matching.

Starting with these 82 seeds and the empty permutation, we use the reflection lemma in strictly increasing order of the target length. The following predecessor sets give all other cases before the uniform tail.

First: 26 reflected lengths below 173.

```
cube     already constructed predecessors m
  8      0
 27      0,7
 64      0,7,19,26
125      0,7,19,26,37,44,56
216      44,56,63,66,68,80,82,85,87,
         100,103,105
```

Together with the 16 exceptional lower seeds and the six algebraic-template lower seeds, these supply 48 distinct lengths below 173.

Second: 49 reflected lengths in $[173, 255] \cup [339, 364]$**.**

```
cube     previously constructed predecessors m
216      0,7,19,26,37
343      0,87,98,100,103,105,110,112,
         115,116,117,118,119,120,121,122,
         123,124,125,126,128,129,130,131,
         132,133,135,147,149,150,151,152,
         155,156,159
512      147,149,150,151,152,155,156,
         159,171
```

The 58 upper exceptional rows, the two upper algebraic-template lengths and these 49 reflections are exactly the 109 integers in $[173, 255] \cup [339, 364]$. There is no circular dependency: each predecessor m is strictly smaller than its constructed length n.

For example, $211 = 343 - 131 - 1$, $340 = 512 - 171 - 1$, and $342 = 343 - 0 - 1$. The first example replaces a previously necessary-looking 53-segment independent row by one suffix reflection.

In particular 48 positive lower lengths and the 109 lengths in $[173, 255] \cup [339, 364]$ are now covered. For $256 \le n \le 338$, let $m = 511 - n \in [173, 255]$, so that $m < n$ and $m + n + 1 = 512 = 8^3$. Lemma 1 then covers all 192 target base lengths. For every $n \ge 365$, the existing cube-gap inequality of Lemma 2 and strong induction extend the permutations to all later lengths. □

The primary proof still contains 74 individual finite integer equalities: the algebraic templates do not establish a completely case-free classification. The audit script `verify_74_plus_templates.py` is optional and reads only the exceptional table, not the original 191 augmenting words or a matching archive. The older 109-row table is preserved as an independent redundant check.

Proposition 2 (A finite parity obstruction at $n = 236$)**.** *The base length* 236 *has cube-sum permutations (as certified by its* 15*-segment atlas row) but admits no cube-sum* involution.

Proof. In the undirected cube-sum graph on $[1, 236]$, remove the following set S of 86 vertices:

$$\begin{aligned} S = {} & [1, 3] \cup [9, 13] \cup [19, 24] \cup [35, 39] \cup [45, 49] \\ & \cup [56, 60] \cup [71, 74] \cup [81, 85] \cup [107, 115] \\ & \cup [117, 121] \cup [126, 130] \cup [136, 140] \\ & \cup [147, 151] \cup [162, 166] \cup [172, 176] \\ & \cup [198, 201] \cup [208, 212], \end{aligned}$$

where $[a, b]$ denotes the integer interval. The only possible edge sums are $8, 27, 64, 125, 216, 343$. Direct substitution into $j = c - i$ shows that $G_{236} - S$ has 87 singleton components, one five-vertex component $\{14, 50, 75, 141, 202\}$, and six even components of sizes $4, 6, 10, 12, 12, 14$. Of the singleton components, only $\{4\}$ admits a loop ($2 \cdot 4 = 8$); the five-vertex component has no loop. Hence there are $86 + 1 = 87$ odd components which admit no internal loop. In any involution, at least one vertex from each such component must be paired with a distinct vertex of S. Since $|S| = 86$, this is impossible. The atlas row gives ordinary permutations, so the involution obstruction does not contradict their existence. □

11 A short, local existence certificate

The preceding reflection argument is infinite and elementary. The finite existence verification can itself be made much more transparent than an array of 192 long permutations: we give *one* explicit involution at size 173, then 191 short alternating-path words. Only 1844 single-digit cube roots are needed.

Lemma 12 (An explicit order-173 involution)**.** *There is a cube-sum permutation on* $[1, 173]$*, given on each listed*

interval I by $p(i) = c - i$. Intervals paired by a double arrow use the same cube c:

intervals I	c
$[1, 7]$	8
$[22, 42]$	64
$[106, 110]$	216
$[8, 12] \longleftrightarrow [15, 19]$	27
$[13, 14] \longleftrightarrow [111, 112]$	125
$[20, 21] \longleftrightarrow [104, 105]$	125
$[43, 103] \longleftrightarrow [113, 173]$	216

Proof. The eleven intervals partition $[1, 173]$. Reflection $i \mapsto c - i$ preserves each of the first three intervals and exchanges the two intervals in each of the last four rows. Thus it defines an involution, and every sum $i + p(i)$ is a cube. □

Lemma 13 (An alternating-word extension)**.** *Suppose p is a cube-sum permutation of $[1, n-1]$. Let $w = r_1 \cdots r_t$ be a finite word of positive integer cube roots. Starting from $u_0 = n$, define $v_j = r_j^3 - u_{j-1}$. For $j < t$, require $v_j \in [1, n-1]$ and let $u_j = p^{-1}(v_j)$. Suppose all these $u_0, u_1, \ldots, u_{t-1}$ are distinct, all $v_1, \ldots, v_t$ are distinct, and $v_t = n$. Then replacing the old matching edges by*

$$q(u_{j-1}) = v_j \quad (1 \le j \le t),$$

while keeping all other old assignments, yields a cube-sum permutation of $[1, n]$.

Proof. The old matching edges removed along this path are $u_j \mapsto v_j$ for $j < t$; these form a bijection between the old visited left and right vertices. Their replacements join u_{j-1} to v_j and have sum r_j^3 by construction. The new left vertex n and the new right vertex n are incorporated, without repeating any endpoint. Every untouched vertex retains its old matching edge. □

The `augmenting_cube_words.txt` appendix lists one word for each $n = 174, \ldots, 364$. Every root is a single decimal digit in $\{2, \ldots, 8\}$. The words are checked using Lemma 13 and the predecessor matching obtained from the previous word; this is *local* verification, not a matching search or a scan to a large cutoff. The independent, standard-library-only `verify_word_paths.py` checks all required edge equalities and endpoint distinctness. The seven-line interval seed plus 1844 root digits certifies all 192 base cases of Theorem 18; combined with the cube-gap induction this proves existence for *every* $n \ge 173$. The older long witness archives remain as secondary independent cross-checks, not as prerequisites of this compact proof.

12 Finite certificates and exact membership

Our finite data are provided as UTF-8 JSON arrays of integers. A valid witness p is checked solely by

$$\mathrm{sort}(p) = (1, \ldots, n), \qquad \forall i \quad i + p_i \in \mathcal{C}.$$

For a negative instance, let $N_n(S) = \{j \in [1, n] : \exists i \in S,\ i + j \in \mathcal{C}\}$. Hall's marriage theorem gives the immediate obstruction

$$|N_n(S)| < |S| \quad \Longrightarrow \quad P(n) = 0. \tag{23}$$

The file `exact_classification_below_173.json` provides 48 explicit valid permutations and 124 explicit Hall-deficient subsets, exactly partitioning $1 \le n \le 172$. Every assertion is directly checked by `verify_extended.py`. For instance, for $n = 172$ the 27 left indices

$$S = \{1, 5, 8, 13, 20, 31, 35, 38, 42, 43, 45, 92, 96, 99,$$
$$104, 111, 122, 126, 129, 133, 134, 136, 153, 157, 160, 165, 172\}$$

have precisely 26 distinct possible right neighbors. Thus $172 \notin \mathcal{G}$, making the threshold below sharp.

For the unbounded tail, only the 109 witness sizes

$$173 \le n \le 255, \qquad 339 \le n \le 364 \tag{24}$$

require explicit stored permutations. Indeed for each $256 \le n \le 338$, put $m = 511 - n \in [173, 255]$; then $m + n + 1 = 512 = 8^3$, so Lemma 1 constructs the missing 83 cases. This compresses the previously saved 192 base cases by $83/192 > 43\%$. All 109 stored witnesses in `core_109.json`, and all 192 reconstructed witnesses, are checked with exact integer arithmetic. The earlier full 192-row file is retained as an additional independent-format audit source.

There is a further useful compression when the 48 verified positive sizes below 173 are reused. An ascending reflection-closure algorithm starts from those small solutions and, at each $173 \le n \le 364$, uses any already certified $m < n$ for which $m+n+1$ is a cube. Precisely 132 lengths are obtained by reflection and only 60 further lengths need explicit witnesses. Here the empty matching is allowed: $n = 342$ follows directly from $c = 343$, with $p(i) = 343 - i$. The 60 additional witnesses are archived in `core_60_with_empty.json`, and `verify_extended.py` certifies this alternate compression independently of the 109-case presentation.

Proposition 3 (Limit of reflection-only compression)**.** *Suppose the empty matching and the* 48 *known positive lengths below* 173 *are provided as initial data. Among proofs of all lengths* $173, \dots, 364$ *that use only the reflection extension of Lemma 1 and independently supplied new full-length seed permutations, at least* 60 *extra seeds are necessary, and* 60 *suffice.*

Proof. Process $n = 173, \dots, 364$ in ascending order. If no cube c yields a previously certified positive predecessor $m = c - n - 1 \in [0, n-1]$, then reflection cannot establish n from earlier data, so a new seed is forced. Otherwise choose such a predecessor and perform reflection, without adding a seed. This deterministic closure rule yields 44 forced seeds in $[173, 255]$, none in $[256, 338]$ (use $c = 512$ and $m = 511 - n$), and 16 in $[339, 364]$. Thus $44 + 16 = 60$, and the attached short integer-only checker reproduces the forced index list. Every forced seed is unavoidable by the definition of the restricted proof rule. □

This is *not* a lower bound on arbitrary mathematical proofs. The alternating-path extension in Lemma 13 is a different operation and indeed replaces those 60 large independent seeds by one explicit involution and a small word table.

Theorem 18 (Complete A006063 membership)**.** *For every integer* $n \ge 173$*,* $P(n) > 0$*. Below* 173*, the complete membership list is*

$$\begin{aligned}\mathcal{G} \cap [1, 172] = \{&7, 19, 26, 37, 44, 56, 63, 66, 68, 80, 82, 85, 87, 98, 100,\\ &103, 105, 110, 112, 115, 116, 117, 118, 119, 120, 121,\\ &122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132,\\ &133, 135, 147, 149, 150, 151, 152, 155, 156, 159, 171\}.\end{aligned}$$

In sequence-index notation this gives

$$\boxed{\text{A006063}(j) = j + 124 \quad (j \ge 49).}$$

Proof. The exact finite witnesses establish $173 \le n \le 364$, and the Hall certificates settle the smaller sizes. For $n \ge 365$, Lemma 2 gives a cube $c \in [n + 174, 2n]$. The predecessor $m = c - n - 1$ satisfies $173 \le m < n$. By strong induction $P(m) > 0$, and Lemma 1 yields $P(n) > 0$. There are precisely 48 positive indices below 173, so the 49th and all later sequence terms are consecutive. □

13 Multiplicity and the second OEIS classification

A cube-sum permutation p is a perfect matching in the bipartite graph with left and right vertices $1, \dots, n$, where (i, j) is an edge exactly when $i + j \in \mathcal{C}$. For any matching p, form a directed graph D_p on the left vertices, putting $i \to j$ whenever $i \ne j$ and (i, p_j) is a cube-sum edge.

Lemma 14 (Alternating-cycle criterion)**.** *For a valid permutation* p*, there exists another distinct valid permutation if and only if* D_p *has a directed cycle.*

Proof. Along a directed cycle $i_1 \to \cdots \to i_t \to i_1$, replace p_{i_s} by $p_{i_{s+1}}$ (cyclic indices). Every moved pair is an allowed edge, and all right vertices remain distinct. Conversely, if $q \ne p$ is another valid permutation, the nontrivial cycles of the permutation $p^{-1} \circ q$ consist of directed cycles in D_p. Thus acyclicity is a rigorous uniqueness certificate. □

The file `second_solutions.json` records directed cycles, not complete duplicate permutations. Direct finite checking establishes: among the 48 positive lengths below 173, exactly

$$E = \{112, 115, 116, 117, 119, 124, 125, 126, 127, 128, 129, 130, 133\}$$

have a directed alternating cycle; every other positive length below 173, as well as 173 and 174, has an acyclic D_p and therefore a unique permutation.

Furthermore, all 190 base cases $175 \le n \le 364$ have *two distinct* alternating-cycle certificates, archived as `second_solutions.json` and `third_solutions.json`. They yield three different permutations, counting the original matching. To extend the multiplicity assertion: for $365 \le n \le 553$, the cube 729 gives $m = 728 - n \in [175, 363]$. The two exceptional lengths $554, 555$ have separate pairs of alternating-cycle certificates obtained by extending the unique solutions at $174, 173$ respectively. For $n \ge 556$, the stronger part of Lemma 2 gives $m = c - n - 1 \ge 175$. Three distinct solutions for m remain distinct after reflection, finishing an induction.

Theorem 19 (Complete A096680 membership)**.** *The positive integers with at least two cube-sum permutations are exactly*

$$\boxed{E \ \cup \ \{175, 176, 177, \dots\}.}$$

Moreover $P(n) \ge 3$ for every $n \ge 175$, while $P(173) = P(174) = 1$. The sequence terms satisfy

$$\boxed{\text{A096680}(j) = j + 161 \quad (j \ge 14).}$$

Sharpness of the constant three. The bound $P(n) \ge 3$ for all $n \ge 175$ cannot be improved to four: $P(175) = 3$ exactly. For the stored matching at 175, the directed alternation graph has precisely one nontrivial strongly connected component,

$$\{4, 11, 16, 23, 41, 48, 102, 114, 168, 175\},$$

and the other 165 vertices are singleton components. Every cycle of a permutation alternative lies in a strongly connected component, so the permanent reduces to the 10-by-10 component's allowed-edge matrix (including the matching self-edges). A direct bitmask dynamic program gives its permanent as 3. This computation is independently reproduced in `verify_extended.py` using exact integers.

14 A nine-gadget exceptional case at $n = 236$

An *involution* is a permutation satisfying $p^2 = \mathrm{id}$. At $n = 236$ there are exactly 512 cube-sum permutations, *none* of which is an involution. This follows from a short alternating-cycle decomposition, not from a general permanent computation.

Define a reference permutation p of $[1, 236]$ by $p(i) = c - i$ on the following fifteen intervals:

I	c	I	c
$[1, 7]$	8	$[8, 9]$	216
$[10, 18]$	64	$[19, 43]$	125
$[44, 45]$	216	$[46, 70]$	125
$[71, 81]$	216	$[82, 106]$	125
$[107, 134]$	343	$[135, 136]$	216
$[137, 170]$	343	$[171, 172]$	216
$[173, 197]$	343	$[198, 208]$	216
$[209, 236]$	343		

These interval images partition $[1, 236]$, and every prescribed sum is a cube.

Lemma 15 (Translation of alternating cycles)**.** *Suppose $p(i) = c_j - i$ on pairwise disjoint integer intervals I_j for $1 \le j \le r$. If $i_j \in I_j$ form a directed alternating cycle $i_1 \to i_2 \to \cdots \to i_r \to i_1$ for p, then every integer shift t with $i_j + t \in I_j$ for all j produces another alternating cycle on $i_j + t$. If t ranges through a set T of such shifts, the translated cycles are disjoint and yield at least $2^{|T|}$ distinct valid cube-sum permutations.*

Proof. Because $p(i_{j+1}+t) = c_{j+1} - i_{j+1} - t$,

$$(i_j + t) + p(i_{j+1} + t) = i_j + c_{j+1} - i_{j+1} = i_j + p(i_{j+1}),$$

so every translated alternating edge has the same cube sum as its original. Since the I_j are pairwise disjoint, different shifts cannot share a vertex within any I_j. The cycles are therefore vertex-disjoint and can be switched independently. □

In the present 236-element template,

$$10 \to 137 \to 46 \to 198 \to 71 \to 10$$

is one alternating 5-cycle, with successive cube sums

$$216, \quad 216, \quad 64, \quad 343, \quad 125.$$

The five vertices belong respectively to the five different reflection blocks $[10, 18]$, $[137, 170]$, $[46, 70]$, $[198, 208]$, $[71, 81]$. The common shifts $t = 0, \ldots, 8$ produce nine disjoint alternating cycles *by pure algebra*, giving immediately

$$\boxed{P(236) \geq 2^9 = 512}$$

without any matching enumeration. The remaining work for the exact count is only to exclude further cycles.

Lemma 16 (Alternating digraph at 236)**.** *Let D_p be the digraph with an arrow $i \to j$ whenever $i \neq j$ and $i + p(j)$ is a cube. Its strongly connected components comprise* 191 *singletons and exactly nine five-vertex sets*

$$S_t = \{10 + t, 46 + t, 71 + t, 137 + t, 198 + t\}, \qquad 0 \leq t \leq 8.$$

Within each S_t the nontrivial arrows form one simple directed 5*-cycle, without chords.*

Proof. This is a finite integer check using the only six possible cubes $8, 27, 64, 125, 216, 343$. The interval formula specifies every edge; the independent `count_236.py` verifier checks all 236 rows, the component partition, and the five-cycle properties with integer arithmetic. It invokes neither a matching solver nor a permanent-computation routine. □

Theorem 20 (Exact count and involution obstruction)**.**

$$\boxed{P(236) = 2^9 = 512.}$$

There is no cube-sum involution on $[1, 236]$.

Proof. Given any other cube-sum permutation q, $f = p^{-1} \circ q$ is a permutation whose arrows lie in D_p or are identity loops. Every cycle of f is confined to one strongly connected component. Each singleton is therefore fixed, while each S_t has exactly two choices: five fixed points or the unique directed 5-cycle. The choices are independent, so there are 2^9 permutations q.

For the central set $S_4 = \{14, 50, 75, 141, 202\}$, the reference permutation restricts to

$$14 \mapsto 50 \mapsto 75 \mapsto 141 \mapsto 202 \mapsto 14.$$

The only two possible restrictions of q to S_4 are this 5-cycle and its inverse, neither of which is an involution. Thus no q is an involution. □

The failure of involutivity at 236 is therefore a specific five-cycle obstruction, not a failure of cube-sum permutation existence. The proof uses only a fifteen-row interval template and an easily audited finite graph-decomposition lemma.

15 A taxicab principle for exponentially many involutions

Positive exponential entropy is not restricted to general permutations with odd directed cycles. Equal sums of two perfect powers generate large families of actual *involutions*. The argument is valid for any fixed exponent $d \geq 2$.

Theorem 21 (Perfect-power taxicab switching)**.** *Fix $d \geq 2$, and suppose positive perfect dth powers A, B, C, D satisfy*

$$\boxed{A + B = C + D, \qquad \frac{3}{2}A \leq C < \frac{1}{2}B.}$$

Let $I_d(n)$ count permutations p of $[1, n]$ with $p^2 = \mathrm{id}$ and every $i + p(i)$ a perfect dth power. For every positive integer u, with $T = u^d$,

$$\boxed{I_d((B - A)T) \geq 2^{\lfloor (AT-1)/2 \rfloor}.} \tag{25}$$

There is an injective encoding of the $\lfloor (AT-1)/2 \rfloor$ binary choices into these involutions which multiplies Hamming distances by exactly four. Consequently,

$$\boxed{\limsup_{n\to\infty} \frac{\log I_d(n)}{n} \geq \frac{A \log 2}{2(B - A)} > 0.}$$

Proof. Since $A < C < B/2$, we have $B > 2A$. Put $N = (B - A)T$ and start with the two-block reflection involution

$$p(i) = \begin{cases} AT - i, & 1 \leq i < AT, \\ BT - i, & AT \leq i \leq N. \end{cases}$$

The two blocks are separately invariant, and the sums are the dth powers $AT = (uA^{1/d})^d$ and $BT = (uB^{1/d})^d$.

Write $H = \lfloor (AT - 1)/2 \rfloor$. For each integer $s = 1, \ldots, H$, consider the four distinct vertices

$$a_s = s, \quad b_s = AT - s, \quad c_s = CT - s, \quad d_s = (B - C)T + s.$$

Here a_s, b_s belong to the first block, while c_s, d_s belong to the second block, since $H \leq AT/2 \leq (C - A)T$. The inequalities $2C < B$ and $2H < AT$ imply that the four coordinate ranges obtained by varying s are pairwise disjoint. Thus all H quadruples are vertex-disjoint.

The reference involution has transpositions $(a_s\, b_s)$ and $(c_s\, d_s)$. For any chosen quadruple, replace these by $(a_s\, c_s)$ and $(b_s\, d_s)$. The new sums are

$$a_s + c_s = CT, \qquad b_s + d_s = (A + B - C)T = DT,$$

both perfect dth powers. This is again an involution, and the independent choices on the H disjoint quadruples give 2^H different legal involutions.

Changing one choice changes precisely the four corresponding permutation values, yielding the exact Hamming distance multiplier four. Finally $H = \frac{1}{2}AT + O(1)$ and $N = (B - A)T$, so taking logarithms along the infinite sequence $u \to \infty$ proves the entropy bound. □

Corollary 5 (Two explicit cubic identities)**.** *For every integer $u \geq 1$, with $T = u^3$,*

$$\boxed{I_3(54406T) \geq 2^{\lfloor (4913T-1)/2 \rfloor}.}$$

This follows from the small identity

$$17^3 + 39^3 = 26^3 + 36^3,$$

using $(A, B, C, D) = (4913, 59319, 17576, 46656)$. A second identity,

$$99^3 + 172^3 = 120^3 + 163^3,$$

gives the stronger optional numerical estimate

$$\boxed{\limsup_{n\to\infty} \frac{\log I_3(n)}{n} \geq \frac{970299 \log 2}{8236298} > 0.}$$

Neither identity is asserted to optimize the constant; the structural taxicab-switching theorem is independent of choosing a special example.

Proof. Each identity and its inequalities $\frac{3}{2}A \le C < B/2$ are verified by integer multiplication, so Theorem 21 applies. In the second case $A = 970299$, $B = 5088448$ and $B - A = 4118149$, producing $A \log 2/[2(B - A)]$ as displayed. □

The construction distinguishes the two entropy sources in the cube-sum graph. Reflection-only descendants of finitely many fixed seeds have zero entropy per site by Corollary 4. Yet globally there are both ordinary permutations *and involutions* with positive upper exponential entropy, by Theorems 22 and 21. The latter needs only a four-term equal-power identity; it does not assume any eventual-existence threshold.

Corollary 6 (Linear entropy with parity-minimal fixed points)**.** *Let $J_0(n)$ count legal fixed-point-free cube-sum involutions on $[1, n]$ for even n, and let $J_1(n)$ count legal cube-sum involutions with exactly one fixed point for odd n. For every* odd *positive integer u, with $T = u^3$,*

$$\boxed{\begin{aligned} J_0(54406T) &\ge 2^{(4913T-1)/2}, \\ J_1(4118149T) &\ge 2^{(970299T-1)/2}. \end{aligned}}$$

In particular, both parity classes admit explicit infinite subsequences on which the number of legal minimally fixed *involutions grows exponentially in their lengths:*

$$\boxed{\begin{aligned} \limsup_{\substack{n\to\infty \\ n \text{ even}}} \frac{\log J_0(n)}{n} &\ge \frac{4913 \log 2}{108812} > 0, \\ \limsup_{\substack{n\to\infty \\ n \text{ odd}}} \frac{\log J_1(n)}{n} &\ge \frac{970299 \log 2}{8236298} > 0. \end{aligned}}$$

Proof. Use the two exact identities from Corollary 5. In Theorem 21, the initial two-block permutation has one fixed point in its first block if and only if AT is even, and one in its second block if and only if BT is even. No taxicab switch can create or remove a fixed point: its four distinct endpoints occur in two transpositions before and after switching, while all unselected positions are unchanged.

For $(a, b) = (17, 39)$ and odd u, both $AT = (au)^3$ and $BT = (bu)^3$ are odd. Thus the baseline and every switched involution are fixed-point-free, proving the even-length bound.

For $(a, b) = (99, 172)$ and odd u, the first cube sum AT is odd, while BT is even, so the second reflection block has exactly one fixed point and the first has none. All switches preserve this single fixed point. The lengths are odd in this case, so one is the minimum possible fixed-point count. The entropy assertions follow on letting odd $u \to \infty$. □

16 A computation-free exponential construction

A particularly simple weighted identity of four cubes produces more permutations than all the previous asymptotic constructions, with *no* finite matching certificate or eventual-existence theorem.

Proposition 4 (Weighted-cube odd-cycle principle)**.** *Let $q \ge 1$, and let A, B, X, Y be positive cubes such that*

$$(q + 1)X + qY = A + 2qB.$$

For $0 \le j \le 2q$ put

$$D_j = \lceil j/2 \rceil \, (B - X) + \lfloor j/2 \rfloor \, (B - Y).$$

Assume $A \le D_j \le B - 2A$ for $j \ge 1$ and $|D_j - D_k| \ge A$ whenever $j \ne k$. Then for every integer $u \ge 1$,

$$\boxed{P((B - A)u^3) \ge 2^{Au^3-1}.}$$

In fact the family encodes $Au^3 - 1$ independent binary choices, each of which changes precisely $2q+1$ permutation positions.

Proof. Put $T = u^3$ and define the two-interval reflection $p(i) = AT - i$ for $i < AT$ and $p(i) = BT - i$ for $AT \le i \le (B-A)T$. The interval hypotheses make this a valid permutation with cube sums AT and BT. For each $1 \le s < AT$, let $v_j = s + D_j T$ for $0 \le j \le 2q$. The gap conditions place v_0 in the first reflection block and the other vertices in the second, while ensuring that the translated cycles for different s are disjoint. The edge sums $v_j + p(v_{j+1})$ are alternately XT and YT for $j < 2q$. The final edge is a cube as well, since

$$v_{2q} + p(v_0) = (A + 2qB - qX - qY)T = XT.$$

Thus the $AT - 1$ cycles can be switched independently, proving the count and Hamming-distance statement. □

For $q = 1$ this recovers the three-cycle identity $2X + Y = A + 2B$. For $q = 2$ it becomes $3X + 2Y = A + 4B$. In particular,

$$3 \cdot 5^3 + 2 \cdot 8^3 = 3^3 + 4 \cdot 7^3$$

gives $(A, B, X, Y) = (27, 343, 125, 512)$ and offsets $(D_0, \dots, D_4) = (0, 218, 49, 267, 98)$, satisfying every hypothesis. Consequently

$$\boxed{P(316u^3) \ge 2^{27u^3 - 1}}$$

for every positive integer u. This unifies the three- and five-cycle constructions through one algebraic identity.

Lemma 17 (Three-cycle identity gadget)**.** *Let A, B, X, Y be positive perfect cubes satisfying*

$$A + 2B = 2X + Y, \qquad A + \tfrac{1}{2}B \le X \le B - A.$$

For every positive integer u, put $T = u^3$ and $N = (B - A)T$. There are at least

$$\boxed{2^{AT-1}}$$

different permutations p of $[1, N]$ for which $i + p(i)$ is a perfect cube.

Proof. The displayed inequalities give $B \ge 4A$. Define a reference permutation by two reflections:

$$p(i) = \begin{cases} AT - i, & 1 \le i < AT, \\ BT - i, & AT \le i \le (B - A)T. \end{cases}$$

The first interval maps to itself under the first reflection, and the second also maps to itself under the second reflection. Every original sum is AT or BT, both perfect cubes.

Set $d_1 = B - X$ and $d_2 = X - A$. The hypotheses imply

$$d_1 \ge A, \qquad d_2 - d_1 \ge A, \qquad d_2 + A = X \le B - A.$$

For every $s = 1, \dots, AT - 1$, the triple

$$C_s = (s,\ s + d_1 T,\ s + d_2 T)$$

lies inside $[1, N]$, with the second and third vertices in the second reflection interval. The three translated ranges are pairwise disjoint by the displayed gap inequalities. Thus all $AT - 1$ triples are pairwise vertex-disjoint.

Along the directed three-cycle $s \to s + d_1 T \to s + d_2 T \to s$, the three alternative matching sums are

$$\begin{aligned} s + p(s + d_1 T) &= XT, \\ (s + d_1 T) + p(s + d_2 T) &= (A + 2B - 2X)T = YT, \\ (s + d_2 T) + p(s) &= XT. \end{aligned}$$

Because T and A, B, X, Y are cubes, every new sum is a cube. Switching any subset of the $AT - 1$ disjoint cycles yields 2^{AT-1} pairwise different legal permutations. □

Theorem 22 (Explicit high-entropy cube-sum permutations)**.** *For every integer $u \ge 1$,*

$$\boxed{P(681707u^3) \ge 2^{148877u^3 - 1}.} \tag{26}$$

Consequently,

$$\boxed{\limsup_{n \to \infty} \frac{\log P(n)}{n} \ge \frac{148877 \log 2}{681707} > 0.} \tag{27}$$

This theorem is entirely computation-free, apart from elementary multiplication of four displayed integers.

Proof. Apply Lemma 17 with

$$A = 53^3 = 148877, \quad B = 94^3 = 830584,$$
$$X = 87^3 = 658503, \quad Y = 79^3 = 493039.$$

Indeed,

$$A + 2B = 1810045 = 2X + Y,$$

and

$$A + \tfrac{1}{2}B = 564169 \le X = 658503 \le 681707 = B - A.$$

Hence the $AT - 1 = 148877u^3 - 1$ explicit cycles are disjoint. More precisely, each has the form

$$s \longrightarrow s + 172081u^3 \longrightarrow s + 509626u^3 \longrightarrow s,$$

with alternative sums $(87u)^3, (79u)^3, (87u)^3$. No matching search, graph decomposition, or auxiliary finite base case is involved. Taking logarithms along the sequence $n = 681707u^3$ proves the second claim. □

Corollary 7 (Linear-distance binary permutation codes)**.** *For $N = 681707u^3$ there is an injective map from $\{0,1\}^{148877u^3-1}$ into legal cube-sum permutations of $[1, N]$ that multiplies all Hamming distances by exactly three. Its distance enumerator is*

$$\big(1 + z^3\big)^{148877u^3-1}.$$

In particular, a greedy binary code subfamily has exponentially many legal permutations with pairwise Hamming distance proportional to N.

Proof. Each bit chooses whether to switch one of the pairwise disjoint three-cycles. A changed bit changes precisely three mapped positions. The distance identity and polynomial follow immediately. The greedy Hamming-ball argument from Corollary 10 applies unchanged. □

17 Power-block lifting and exponential growth on a subsequence

The nine disjoint alternating cycles at 236 can be amplified by a purely algebraic interval construction, yielding exponentially many solutions at infinitely many *different* lengths. The lifting mechanism works for every perfect-power exponent, not only cubes.

Lemma 18 (Independent-fiber perfect-power tensorization)**.** *Fix $d \ge 2$ and $M \ge 1$, and suppose $P_d(M) > 0$. For any odd positive integer t put $L = t^d$ and $A = (L-1)/2$. For each $r \in [1, L]$ choose independently a dth-power-sum permutation p_r of $[1, M]$. Define a permutation Q on the interval $I = [A + 1, A + ML]$ by*

$$Q\big(A + (i-1)L + r\big) = A + (p_r(i) - 1)L + (L + 1 - r). \tag{28}$$

Then every $x + Q(x)$ is a perfect dth power, and the L independent choices give exactly

$$\boxed{P_d(M)^L}$$

different permutations of I.

If instead all fibers use the same p possessing h pairwise vertex-disjoint directed alternating cycles, then Q has at least hL pairwise vertex-disjoint directed alternating cycles.

Proof. The input positions have unique coordinates (i, r). For fixed r, the map $i \mapsto p_r(i)$ is a permutation of $[1, M]$, while the output residue $L + 1 - r$ identifies the original fiber r uniquely. Thus (28) is a bijection. For $x = A + (i-1)L + r$,

$$\begin{aligned} x + Q(x) &= 2A + L(i + p_r(i) - 1) + 1 \\ &= L(i + p_r(i)). \end{aligned}$$

If $i + p_r(i) = k_{i,r}^d$, then this sum equals $(tk_{i,r})^d$. The map from the tuple $(p_1, \dots, p_L)$ to Q is injective, since all $p_r(i)$ are recovered from the output block indices; hence there are exactly $P_d(M)^L$ such interval permutations.

Finally, when every $p_r = p$, an alternating edge $i \to j$ of p gives $x_{i,r} + Q(x_{j,r}) = L(i + p(j))$ for each $x_{i,r} = A + (i-1)L + r$, again a perfect power. Every original alternating cycle consequently lifts to one cycle in each fiber, and disjoint fibers and original cycles remain disjoint. □

Proposition 5 (Inverse-paired fiber involutions)**.** *In the setting of Lemma 18, let $I_d(M)$ be the number of valid power-sum involutions of length M. The exact number of lifted interval permutations of the form* (28) *which themselves are involutions is*

$$\boxed{I_d(M)\,P_d(M)^{(L-1)/2}.} \tag{29}$$

Proof. Write $r' = L+1-r$. Since (28) sends (i,r) to $(p_r(i), r')$, its square sends (i,r) to $(p_{r'}(p_r(i)), r)$. Thus the lifted interval permutation is an involution if and only if

$$p_{L+1-r} = p_r^{-1} \qquad (1 \le r \le L).$$

The inverse of any valid power-sum permutation is valid, by symmetry of the condition $i + p(i)$. The $(L-1)/2$ pairs of distinct fibers may therefore be chosen independently in $P_d(M)$ ways each; on the unique central fiber $r = (L+1)/2$, the chosen permutation must be an involution and has $I_d(M)$ choices. Injectivity of the fiber encoding gives (29). □

Corollary 8 (Transfer from general multiplicity to involution entropy)**.** *Fix $d \ge 2$. Suppose valid dth-power-sum involutions exist for every sufficiently large length. If some M satisfies $I_d(M) > 0$ and $P_d(M) \ge 2$, then*

$$\boxed{\limsup_{n\to\infty} \frac{\log I_d(n)}{n} \ge \frac{\log P_d(M)}{2M+1} > 0.}$$

Proof. For sufficiently large odd t, put $L = t^d$, $A = (L-1)/2$ and $B = A + ML$. Choose a valid involution on the prefix $[1, A]$. Proposition 5 constructs at least $P_d(M)^{(L-1)/2}$ valid involutions on the middle interval. For the suffix choose the least perfect dth power $C = k^d > (2M+1)L$, and reflect $[B+1, C-B-1]$ through C. This is a legal involution because $B + (C-B-1) + 1 = C$. The resulting full size is

$$n = C - B - 1 = \big(M + \tfrac{1}{2}\big)L + O(t^{d-1}),$$

since $k = (2M+1)^{1/d}t + O(1)$. The prefix and suffix are fixed across all middle choices. Taking logarithms and dividing by n gives the result. □

This principle shows that power-sum involution families may have positive linear entropy even when the independent local choices on the middle blocks are *not themselves involutions*: inverse-pairing complementary fibers is enough, provided one central involution exists. In the cubic case the explicit 316-element two-reflection construction supplies such a central involution and several alternative ordinary permutations. The direct taxicab-switching theorem in Section 15 does not require this eventual prefix-existence hypothesis.

Theorem 23 (An exponential subsequence)**.** *For every odd integer $t \ge 9$, define*

$$\boxed{n_t = \frac{551t^3 - 1}{2}.}$$

Then

$$\boxed{P(n_t) \ge P\left(\frac{t^3-1}{2}\right) 512^{t^3} \ge 3 \cdot 2^{9t^3}.}$$

In particular,

$$\boxed{\limsup_{n\to\infty} \frac{\log P(n)}{n} \ge \frac{18\log 2}{551} > 0.}$$

Proof. Take the explicit $M = 236$ permutation from Section 14, possessing $h = 9$ disjoint alternating 5-cycles. Let $L = t^3$, $A = (L-1)/2$, and $B = A + 236L = (473L-1)/2$. Because $t \ge 9$, $A \ge 364 \ge 173$, so Theorem 18 supplies a valid permutation on the prefix $[1, A]$. By Theorem 19, $P(A) \ge 3$. The exact count $P(236) = 512$ from Theorem 20 and Lemma 18 provide 512^L distinct valid middle-block permutations on $[A+1, B]$, without needing to select a preferred alternating-cycle orientation.

Now let $C = (8t)^3 = 512L$ and set

$$n_t = C - B - 1 = \frac{551L - 1}{2}.$$

The final suffix $[B+1, n_t]$ has exactly $39L > 0$ elements. Define the permutation there by $x \mapsto C - x$. It permutes that suffix because $C = B + n_t + 1$, and every new sum is the cube C. Combining the three disjoint intervals produces a cube-sum permutation on $[1, n_t]$. The prefix and middle-block choices are independent, while the suffix is fixed. Consequently at least $P(A)\,512^L$ distinct permutations are obtained, and $P(A) \geq 3$ completes the proof. The older $9L$ independent-cycle construction is a special subfamily of the new tensorized construction. □

Theorem 24 (Asymptotically optimal terminal reflection)**.** *Let $t \geq 9$ be odd and $L = t^3$. Let k_t be the least integer satisfying*

$$k_t^3 > 473L,$$

and define

$$\boxed{N_t = k_t^3 - \frac{473L+1}{2}.}$$

Then

$$\boxed{P(N_t) \geq P\left(\frac{L-1}{2}\right) 512^L \geq 3 \cdot 2^{9L}.}$$

Furthermore,

$$N_t = \left(\frac{473}{2} + o(1)\right) L \qquad (t \to \infty),$$

and therefore

$$\boxed{\limsup_{n\to\infty} \frac{\log P(n)}{n} \geq \frac{18 \log 2}{473}.} \tag{30}$$

Proof. Take the 236-element tensorized gadget of Lemma 18. The valid prefix has length $A = (L-1)/2$ and the lifted middle block ends at $B = (473L-1)/2$. The strict inequality $k_t^3 > 473L = 2B+1$ guarantees the suffix interval $[B+1, N_t]$ is nonempty: indeed

$$N_t - B = k_t^3 - 473L > 0.$$

Reflection through the cube k_t^3 permutes that suffix since $B + N_t + 1 = k_t^3$. The middle blocks have 512^L independent choices, the prefix has $P(A)$ choices, and the suffix is fixed. The first bound follows, and $P(A) \geq 3$ because $A \geq 364$. By minimality,

$$(k_t - 1)^3 \leq 473t^3 < k_t^3,$$

so $k_t = 473^{1/3}t + O(1)$. Thus $k_t^3 = 473t^3 + O(t^2)$ and $N_t = (473/2)t^3 + O(t^2)$. Dividing $9L \log 2$ by N_t establishes (30). □

The fixed terminal cube $(8t)^3$ used in Theorem 23 is convenient for a simple closed formula $n_t = (551t^3 - 1)/2$, but is not asymptotically optimal. Selecting the *least* admissible terminal cube as above shrinks the suffix from $39t^3$ to $O(t^2)$, and improves the linear entropy coefficient from $18 \log 2/551$ to $18 \log 2/473$.

Corollary 9 (Multiplicity forces exponential subsequences)**.** *Fix $d \geq 2$. Suppose $P_d(n) > 0$ for all sufficiently large n and that $P_d(M) \geq 2$ for some $M \geq 1$. Choose any integer b with $b^d > 2M + 1$. For all sufficiently large odd t, define*

$$n_t = \left(b^d - M - \tfrac{1}{2}\right)t^d - \tfrac{1}{2}.$$

Then

$$P_d(n_t) \geq P_d\left(\frac{t^d - 1}{2}\right) P_d(M)^{t^d},$$

and in particular

$$\boxed{\limsup_{n\to\infty} \frac{\log P_d(n)}{n} \geq \frac{\log P_d(M)}{b^d - M - \frac{1}{2}} > 0.}$$

Proof. Let $L = t^d$, $A = (L-1)/2$ and $B = A + ML$. Eventual existence supplies a valid prefix on $[1, A]$ for all sufficiently large odd t, with $P_d(A)$ possible choices. Lemma 18 gives $P_d(M)^L$ independent middle-block choices. Now put $C = (bt)^d = b^d L$ and $n_t = C - B - 1$. Since $b^d > 2M + 1$, the suffix $[B+1, n_t]$ is nonempty; reflection $x \mapsto C - x$ permutes it and makes every sum equal to C. The three intervals are invariant under the constructed permutation, and their choices are independent. Taking logarithms and dividing by n_t gives the bound. □

Theorem 25 (An exact binary cube-sum permutation code)**.** *For each odd $t \geq 9$, put $L = t^3$ and $n_t = (551L - 1)/2$. There is an injective map*

$$\Phi_t : \{0,1\}^{9L} \longrightarrow \{\textit{cube-sum permutations of } [1, n_t]\}$$

such that for all bit strings u, v,

$$\boxed{d_{\mathrm{H}}(\Phi_t(u), \Phi_t(v)) = 5\, d_{\mathrm{H}}(u, v).} \tag{31}$$

In particular, relative to $\Phi_t(0)$, the exact Hamming-distance enumerator of this explicit family is

$$\boxed{\sum_{u \in \{0,1\}^{9L}} z^{d_{\mathrm{H}}(\Phi_t(u), \Phi_t(0))} = (1 + z^5)^{9L}.}$$

Proof. The proof of Theorem 20 identifies every 236-element cube-sum permutation with a unique nine-bit vector recording which of the nine disjoint directed five-cycles have been switched. Two different choices on one five-cycle change all five mapped values, while all untouched components agree. Thus their permutation Hamming distance is exactly five times the bitwise Hamming distance.

Take the L independent base-permutation fibers of Lemma 18, and concatenate their nine-bit selectors into $u \in \{0,1\}^{9L}$. Different fibers occupy disjoint input positions, so distances add exactly. Finally append a fixed valid prefix and the fixed cube-reflection suffix of Theorem 23. Neither changes the Hamming distances. The generating polynomial follows because each of the $9L$ bits independently contributes either distance zero or distance five. □

Corollary 10 (Positive-rate, linear-distance families)**.** *Let $H_2(x) = -x \log_2 x - (1 - x) \log_2(1 - x)$ be binary entropy. For every odd $t \geq 9$, there is a subfamily $\mathcal{F}_t$ of cube-sum permutations on $[1, n_t]$ such that*

$$|\mathcal{F}_t| \geq 2^{\,9L(1 - H_2(1/4))}, \qquad d_{\min}(\mathcal{F}_t) > \frac{45L}{4}.$$

Consequently both $\log |\mathcal{F}_t| = \Omega(n_t)$ and $d_{\min}(\mathcal{F}_t) = \Omega(n_t)$.

Proof. Greedily select a binary word of length $N = 9L$, and delete all words within Hamming distance $\lfloor N/4 \rfloor$ of the selected word. A binary Hamming ball of this radius has at most $2^{NH_2(1/4)}$ members, so the greedy packing yields at least $2^{N(1 - H_2(1/4))}$ codewords, with pairwise binary distance exceeding $N/4$. Applying Φ_t multiplies all distances by exactly five, while preserving cardinality. Since $n_t = (551L - 1)/2$, both claimed linear asymptotics follow. □

Corollary 11 (Optimal general terminal choice)**.** *Under the assumptions of Corollary 9, the stronger universal estimate holds:*

$$\boxed{\limsup_{n \to \infty} \frac{\log P_d(n)}{n} \geq \frac{\log P_d(M)}{M + \frac{1}{2}}.}$$

Proof. Put $L = t^d$ with odd t and $B = ((2M + 1)L - 1)/2$. Take the least integer k for which $k^d > (2M + 1)L$. Since $k = (2M+1)^{1/d}t + O(1)$, the reflected suffix gives a terminal size $n = k^d - B - 1$ with $n = (M + \frac{1}{2})L + O(t^{d-1})$. The independent-fiber middle and eventually valid prefix yield $P_d(n) \geq P_d(M)^L$. Taking logarithms proves the bound. □

Theorem 26 (Power-divisor tensor recurrence)**.** *Fix an exponent $d \geq 2$ and a nonnegative integer n. For any odd positive integer t such that $L = t^d$ divides $2n + 1$, set*

$$A = \frac{L - 1}{2}, \qquad q = \frac{2n + 1}{L}, \qquad M_b = b^d - \frac{q + 1}{2}.$$

With $P_d(0) = 1$, we have

$$\boxed{P_d(n) \geq P_d(A) \sum_{\substack{b \geq 2,\ M_b \geq 0 \\ b^d < q}} P_d(M_b)^{\,L}.} \tag{32}$$

For $t = 1$, this is exactly the earlier one-step interval-reflection counting recurrence (now also allowing the empty prefix).

Proof. For each summand choose one permutation on $[1, A]$ and, independently, L permutations of $[1, M_b]$. Apply Lemma 18 on $[A+1, B]$, where $B = A + M_b L$. The equation defining M_b gives

$$B + n + 1 = b^d L = (bt)^d, \qquad n - B = (q - b^d)L > 0.$$

Thus the final interval $[B+1, n]$ is permuted by the reflection $x \mapsto (bt)^d - x$. All three pieces are valid and disjoint, producing $P_d(A)P_d(M_b)^L$ distinct permutations. Different b have different last entries $p(n) = (bt)^d - n$, so these families are disjoint and can be added. When $t = 1$, $L = 1$, $A = 0$ and $M_b = b^d - n - 1$, recovering the reflection recurrence. □

This formula gives a precise arithmetic description of the advantage and limitation of the tensor construction: a nontrivial block scale $t > 1$ requires the cubic divisibility condition $t^d \mid (2n+1)$. It does not, by itself, prove exponential growth at *every* length.

Proposition 6 (Arithmetic sparsity of nontrivial block lifting)**.** *The set of positive integers n for which some odd integer $t \geq 3$ satisfies $t^3 \mid (2n+1)$ has natural density*

$$\boxed{1 - \frac{8}{7\zeta(3)} = 0.0492487\ldots.}$$

Thus more than 95% of lengths have no nontrivial integral cube scale for the particular centered tensor construction used in Theorem 26.

Proof. An odd integer is cube-free precisely when it is divisible by no prime cube p^3, with p odd. By Möbius inclusion–exclusion, the number of positive $n \leq X$ for which $2n+1$ is cube-free equals

$$\begin{aligned}
\sum_{\substack{n \leq X \\ 2n+1 \text{ cube-free}}} 1 &= \sum_{\substack{d \leq (2X+1)^{1/3} \\ d \text{ odd}}} \mu(d) \#\{n \leq X : d^3 \mid 2n+1\} \\
&= X \sum_{\substack{d \geq 1 \\ d \text{ odd}}} \frac{\mu(d)}{d^3} + O(X^{1/3}) \\
&= X \prod_{p \text{ odd}} (1 - p^{-3}) + O(X^{1/3}) \\
&= \frac{8X}{7\zeta(3)} + O(X^{1/3}).
\end{aligned}$$

Every odd composite $t \geq 3$ has an odd prime divisor, so $t^3 \mid 2n+1$ is possible for some $t \geq 3$ exactly when $2n+1$ is not cube-free. Taking complements proves the claim. □

This is a limitation of the *specific centered tensor construction*, not an upper bound on $P(n)$ or a no-go theorem for other embeddings. In particular, the exponential-in-n subsequence result cannot by itself be promoted to a uniform exponential lower bound using the same integral scales for every n.

Lemma 19 (A denser finite alternating-cycle gadget)**.** *There exists a cube-sum permutation of $[1, 362]$ admitting at least 40 pairwise vertex-disjoint alternating cycles. Consequently,*

$$\boxed{P(362) \geq 2^{40}.}$$

Proof. An explicit certificate in `certificates/forty_cycles_362.json` describes a valid permutation using 43 integer interval-reflection rows $p(i) = c - i$ and lists 40 alternating directed cycles on 182 distinct vertices. Each listed edge is checked by the single integer identity $i + p(j) = k^3$. These cycles are vertex-disjoint and can be switched independently. The standalone `verify_362.py` checker directly verifies every row and edge without any matching solver or permanent algorithm. □

Theorem 27 (Improved explicit exponential subsequence)**.** *For every odd integer $t \geq 7$, let*

$$N_t = \frac{733t^3 - 1}{2}.$$

Then

$$\boxed{P(N_t) \geq P\left(\frac{t^3-1}{2}\right) 2^{40t^3} \geq 2^{40t^3}.}$$

In particular,

$$\boxed{\limsup_{n\to\infty} \frac{\log P(n)}{n} \geq \frac{80\log 2}{733} > 0.0756.}$$

Proof. In Theorem 26 take $d = 3$, $M_b = 362$, $b = 9$. Then $b^3 = 729$ and $q = 2 \cdot 729 - 2 \cdot 362 - 1 = 733$, so $b^3 < q$ and $n = N_t$. At $t = 7$ the prefix length $A = (7^3 - 1)/2 = 171$ is a known positive index from the finite classification. For every larger odd t, $A \geq 364$, so the existence theorem applies. Hence $P(A) > 0$ and

$$P(N_t) \geq P(A)P(362)^{t^3} \geq 2^{40t^3}.$$

Since $N_t \sim (733/2)t^3$, division by N_t gives the stated exponential coefficient. □

The constant here is stronger than the earlier 236-seed coefficient $18\log 2/551$. It is a rigorously certified *lower bound*, not a claim of optimal asymptotic entropy.

This gives an *exponential limsup*, not an exponential lower bound for all n. In particular, it is logically distinct from the uniform, near-critical reflection estimates in Section 19.

18 Many permutations: an exponential cube-root bound

Besides qualitative multiplicity, the reflection family yields a strong quantitative statement. Let $K(n) = \lfloor \sqrt[3]{2n} \rfloor$, with the floor defined using integer comparisons (not floating point).

Lemma 20 (Three disjoint embeddings)**.** *If $K(n) = k \geq 27$, each of $k^3, (k-1)^3, (k-2)^3$ gives a reflection extension from a predecessor $m \geq 173$ with $m < n$, and all three predecessors obey $K(m) \geq k - 6$.*

Proof. By definition, $k^3 \leq 2n < (k+1)^3$. For the smallest of these cubes, set $m = (k-2)^3 - n - 1$. Then

$$2m > 2(k-2)^3 - (k+1)^3 - 2 = k^3 - 15k^2 + 21k - 19.$$

The right side exceeds $(k-6)^3$ for $k \geq 27$, since the difference is

$$3k^2 - 87k + 197 > 0 \quad (k \geq 27).$$

Thus $2m > (k-6)^3 \geq 21^3$, which implies $m \geq 173$ and $K(m) \geq k - 6$. Also $(k-2)^3 < k^3 \leq 2n$, so $m < n$. The other two cubes give larger predecessors and satisfy the same bounds. Distinct cubes yield disjoint matching families by Lemma 1. □

Theorem 28 (A096904 quantitative bound)**.** *For all integers $n \geq 173$, with $q(n) = \max\{0, \lfloor (K(n) - 27)/6 \rfloor\}$,*

$$\boxed{P(n) \geq 3^{q(n)}}.$$

In particular

$$\log P(n) \;\geq\; \frac{\log 3}{6}(2n)^{1/3} - O(1),$$

so $P(n)$ grows at least exponentially in $n^{1/3}$.

Proof. If $K(n) < 27$ then $q(n) = 0$, and Theorem 18 supplies $P(n) \geq 1$. For $K(n) \geq 27$, Lemma 20 provides three distinct extensions from smaller predecessors $m_i \geq 173$ with $K(m_i) \geq K(n) - 6$. Thus $q(m_i) \geq q(n) - 1$. By strong induction and the disjointness of the three extension images,

$$P(n) \geq \sum_{i=1}^{3} P(m_i) \geq 3 \cdot 3^{\max(0, q(n)-1)} \geq 3^{q(n)}.$$

The displayed logarithmic asymptotic follows immediately. □

19 A universal reflection-entropy bound

The fixed three-branch construction above is only one instance of a general branching mechanism. In fact, one may use *any fixed number* of the largest available cubes and obtain a sharp limiting coefficient for this family of lower bounds.

For each integer $s \geq 2$, let $\rho_s \in (1,2)$ be the unique solution of

$$\sum_{h=1}^{s} \rho_s^{-h} = 1, \qquad B_s = 2s^2 + 4s - 1. \tag{33}$$

Let $K(n) = \lfloor \sqrt[3]{2n} \rfloor$ as before.

Theorem 29 (Reflection entropy)**.** *For all $s \geq 2$ and $n \geq 173$,*

$$\boxed{P(n) \geq \rho_s^{(K(n)-B_s)/2}.} \tag{34}$$

In particular,

$$\boxed{\liminf_{n\to\infty} \frac{\log P(n)}{(2n)^{1/3}} \;\geq\; \frac{\log 2}{2}.} \tag{35}$$

Equivalently, for every $\varepsilon > 0$ there is a constant C_ε such that

$$P(n) \geq \exp\Big(\big(\tfrac{1}{2}\log 2 - \varepsilon\big)\,(2n)^{1/3} - C_\varepsilon\Big) \quad (n \geq 173).$$

Proof. Fix s and write $B = B_s$, $\rho = \rho_s$. We prove (34) by strong induction. For $K(n) < B$, the right side is at most 1, and $P(n) \geq 1$ by Theorem 18.

Now suppose $k = K(n) \geq B$. For each $j = 0, \ldots, s-1$, take the cube $c_j = (k-j)^3$ and put $m_j = c_j - n - 1$. Since $k^3 \leq 2n < (k+1)^3$, the following exact polynomial inequality holds:

$$\begin{aligned} 2m_j - (k-2j-2)^3 &\geq 2(k-j)^3 - (k+1)^3 - 1 - (k-2j-2)^3 \\ &= 3k^2 - (6j^2+24j+15)k \\ &\qquad + 6j^3 + 24j^2 + 24j + 6 \\ &\geq 0. \end{aligned}$$

Dividing the final polynomial by 3, its coefficient of $-k$ is $2j^2+8j+5$ and its constant term is positive. Hence the last inequality follows from $k \geq B \geq 2j^2+8j+5$. Also $k - 2j - 2 \geq k - 2s \geq 8$, so $m_j \geq 256 \geq 173$ and $K(m_j) \geq k - 2(j+1)$. Each $c_j \leq k^3 \leq 2n$, so $m_j < n$.

The s reflection extensions are pairwise disjoint by their distinct last images, as in Lemma 1. Using the strong induction hypothesis, we obtain

$$\begin{aligned} P(n) &\geq \sum_{j=0}^{s-1} P(m_j) \\ &\geq \sum_{j=0}^{s-1} \rho^{(K(m_j)-B)/2} \\ &\geq \rho^{(k-B)/2} \sum_{j=0}^{s-1} \rho^{-(j+1)} = \rho^{(k-B)/2}. \end{aligned}$$

This proves the first assertion. The ρ_s are strictly increasing and converge to 2, since their defining finite geometric sums approach $\sum_{h\geq 1} 2^{-h} = 1$. For each fixed s, B_s is a constant independent of n and $K(n) = (2n)^{1/3} + O(1)$. Therefore the ratio on the left side of (35) is at least $\frac{1}{2}\log\rho_s$ in the limit. Letting $s \to \infty$ proves (35). □

Corollary 12 (Other power sums, conditional on existence)**.** *For an integer $d \geq 2$, let $P_d(n)$ count permutations of $[1,n]$ in which every $i + p_i$ is a perfect dth power. If $P_d(n) > 0$ for all sufficiently large n, then*

$$\liminf_{n\to\infty} \frac{\log P_d(n)}{(2n)^{1/d}} \geq \frac{\log 2}{2}.$$

Proof. For any fixed s and $j = 0, \dots, s-1$, write $k = \lfloor (2n)^{1/d} \rfloor$ and use $c_j = (k-j)^d$, $m_j = c_j - n - 1$. The binomial theorem gives, uniformly over these finitely many j,

$$2(k-j)^d - (k+1)^d - (k-2j-2)^d = dk^{d-1} + O_{d,s}(k^{d-2}) > 1$$

for all sufficiently large k. It follows that m_j is a smaller positive length in the eventual-existence range and $\lfloor (2m_j)^{1/d} \rfloor \geq k - 2j - 2$. The distinct reflection families obey the same s-branch induction as Theorem 29, with a suitable constant initial offset in k. The ρ_s tend to 2 as before. □

Theorem 30 (A near-critical lower bound with explicit loss)**.** *For $K(n) = \lfloor \sqrt[3]{2n} \rfloor$ and every $n \geq 173$,*

$$\boxed{P(n) \geq \frac{127}{K(n)}\, 2^{(K(n)-127)/2}.} \tag{36}$$

More generally, for every real $\varepsilon > 0$ there exists $C_\varepsilon > 0$ such that

$$\boxed{P(n) \geq C_\varepsilon \frac{2^{K(n)/2}}{K(n)^\varepsilon} \qquad (n \geq 173).} \tag{37}$$

Proof. Write $k = K(n)$. The exact cubic polynomial inequality in the proof of Theorem 29 works whenever

$$k \geq B_s = 2s^2 + 4s - 1, \qquad s \geq 2;$$

it supplies s *disjoint* reflection embeddings with predecessor indices $m_h \in [173, n-1]$ and $K(m_h) \geq k - 2h$ for $h = 1, \dots, s$.

First set $F(k) = 2^{k/2}/k$, which is increasing for all integer $k \geq 3$. For $k \geq 128$ choose $s = \lfloor \log_2 k \rfloor$. Then $s \geq 7$ and $k \geq 2^s \geq B_s$: the last inequality holds at $s = 7$, since $128 \geq 125$, and is preserved upon doubling, because $2B_s \geq B_{s+1}$ for all $s \geq 7$. Moreover $k - 2s \geq 8$, so every predecessor is in the increasing range of F. By the reflection counting lemma,

$$\begin{aligned} \frac{\sum_{h=1}^s F(k-2h)}{F(k)} &= \sum_{h=1}^s \frac{2^{-h}k}{k-2h} \\ &\geq \sum_{h=1}^s 2^{-h}\left(1 + \frac{2h}{k}\right) \\ &\geq 1 - 2^{-s} + \frac{2}{k} > 1, \end{aligned}$$

using $\sum_{h=1}^s h2^{-h} \geq 1$ for $s \geq 2$ and $k < 2^{s+1}$. For $k < 128$, Theorem 18 ensures $P(n) \geq 1$. Because F is increasing for $k \geq 7$, this implies $P(n) \geq F(k)/F(127)$. Strong induction using the above inequality extends the same bound to all $k \geq 128$. Since $F(127) = 2^{127/2}/127$, this is exactly (36).

For the more general result, fix $\varepsilon > 0$ and define $F_\varepsilon(k) = 2^{k/2}/k^\varepsilon$. It is increasing for all sufficiently large k. For sufficiently large k, set $s = \lceil \log_2(k/\varepsilon) \rceil$, so $s = O_\varepsilon(\log k)$, $k \geq B_s$ and $2^{-s} \leq \varepsilon/k$. Each predecessor is in the increasing range of F_ε. By the elementary Bernoulli inequality,

$$\left(\frac{k}{k-2h}\right)^\varepsilon \geq 1 + \frac{2\varepsilon h}{k}.$$

Consequently

$$\begin{aligned} \frac{\sum_{h=1}^s F_\varepsilon(k-2h)}{F_\varepsilon(k)} &\geq 1 - 2^{-s} + \frac{2\varepsilon}{k}\sum_{h=1}^s h2^{-h} \\ &\geq 1 - 2^{-s} + \frac{2\varepsilon}{k} > 1. \end{aligned}$$

Choose an initial K_ε large enough for all these inequalities to hold at $k \geq K_\varepsilon$. By positivity, the bound $P(n) \geq C_\varepsilon F_\varepsilon(k)$ holds for $k < K_\varepsilon$ on taking $C_\varepsilon = 2^{-K_\varepsilon/2}$. The same strong induction proves it for every larger k, yielding (37). □

These bounds approach the *critical* reflection coefficient $\frac{1}{2}\log 2$ with a logarithmic (indeed arbitrarily small polynomial) loss. Their proofs use no new computer-assisted certificate beyond the existence theorem.

The coefficient $(\log 2)/2$ is a *proved lower bound*, not an asserted exact exponential growth rate. The argument needs only the qualitative existence Theorem 18; it makes no further use of numerical matching data.

20 An efficient constructive algorithm

The proof is also algorithmic. Given $n > 364$, choose the *least* cube $c = k^3 \geq n + 174$ and put $m = c - n - 1$. The cube-gap lemma ensures $173 \leq m < n$. Minimality gives

$$(k-1)^3 < n + 174 \leq k^3 \quad \Longrightarrow \quad 173 \leq m \leq 3k^2 - 3k + 173 = O(n^{2/3}).$$

Repeat until the remaining size is in the certified interval $173 \leq m \leq 364$, then fill each recorded interval by the explicit reflection $p_i = c - i$.

Proposition 7 (Linear-output construction)**.** *A valid cube-sum permutation of any $n \geq 173$ can be constructed in $O(n)$ unit-cost arithmetic operations, with $O(\log \log n)$ reflection layers.*

Proof. The predecessor bound $m = O(n^{2/3})$ implies $\log m \leq (2/3) \log n + O(1)$, so the number of iterations before the fixed-size base is $O(\log \log n)$. The returned intervals are disjoint and their union, together with the base interval, is $\{1, \ldots, n\}$. Therefore exactly n entries are written. Each exact integer cube root can be computed using $O(\log n)$ comparisons, which contributes $O(\log n \log \log n) = o(n)$ additional unit-cost operations. □

The accompanying `construct_linear.py` implements this procedure with exact integer cube roots, and can also display the recursion plan without allocating the full output.

21 Reproduction and limitations

All programs require only Python 3 standard-library modules:

```
python papers/A006063/verify.py
python papers/A006063/independent_match.py
python papers/A006063/verify_extended.py
python papers/A006063/construct_linear.py
python papers/A006063/verify_word_paths.py
python papers/A006063/count_236.py
```

The archived data consist of 109 core witnesses, the legacy 192-row all-base cross-check, 48 small positive permutations, 124 negative Hall sets, and compact pairs of alternating-cycle certificates. The infinite inductive and counting lemmas are mathematical arguments rather than extrapolations from numerical experiments.

The status is a *computer-assisted proof candidate* pending independent scrutiny of the finite certificate, its interpretation, and literature priority. The present manuscript does not compute the exact values of $P(n)$ in general, nor does it claim external peer review.

A Finite interval-reflection atlas

Each row n lists pairs $b : k$: the consecutive input interval ending at b is mapped by $p(i) = k^3 - i$. The first interval starts at 1. There are 74 exceptional rows and 969 segments. Three parameterized block families prove eight additional base lengths. For every row, Lemma 9 reduces checking the permutation to equality of two lists of integers.

82: 2:4 5:2 21:3 42:4 61:5 63:4 82:5

116: 2:4 3:3 4:2 8:3 9:5 10:3 11:5 15:3 16:5 17:3 18:5 22:3 23:5 24:3 61:5 63:4 100:5 101:6 102:5 106:6 107:5 108:6 109:5 113:6 114:5 115:6 116:5

118: 1:3 2:2 3:3 4:2 5:3 6:2 7:5 8:3 9:5 10:3 11:5 12:3 14:5 15:3 16:5 17:3 18:5 19:3 21:5 22:3 23:5 24:3 25:5 26:3 98:5 99:6 100:5 101:6 102:5 103:6 105:5 106:6 107:5 108:6 109:5 110:6 112:5 113:6 114:5 115:6 116:5 117:6 118:5

120: 2:4 3:3 4:2 9:5 10:3 16:5 17:3 23:5 24:3 61:5 63:4 100:5 101:6 107:5 108:6 114:5 115:6 120:5

121: 2:4 3:3 9:5 10:3 16:5 17:3 23:5 24:3 61:5 63:4 100:5 101:6 107:5 108:6 114:5 115:6 121:5

123: 2:4 31:5 32:4 61:5 63:4 92:5 93:6 122:5 123:6

125: 3:5 4:2 29:5 30:4 33:5 34:4 90:5 91:6 94:5 95:6 120:5 121:6 124:5 125:6

126: 2:4 5:2 28:5 35:4 61:5 63:4 89:5 96:6 119:5 126:6

127: 1:5 4:2 5:3 6:2 11:5 12:3 18:5 19:3 25:5 26:3 27:5 30:4 31:5 36:4 88:5 91:6 92:5 97:6 102:5 103:6 109:5 110:6 116:5 117:6 118:5 121:6 122:5 127:6

129: 3:5 4:2 8:3 11:5 15:3 18:5 22:3 25:5 30:4 33:5 38:4 86:5 91:6 94:5 99:6 102:5 106:6 109:5 113:6 116:5 121:6 124:5 129:6

131: 3:5 23:3 30:4 33:5 40:4 84:5 91:6 94:5 121:6 124:5 131:6

132: 3:5 4:2 22:3 30:4 33:5 41:4 83:5 91:6 94:5 121:6 124:5 132:6

150: 3:5 4:2 30:4 33:5 59:4 65:5 91:6 94:5 121:6 124:5 150:6

151: 3:5 30:4 33:5 60:4 64:5 91:6 94:5 121:6 124:5 151:6

155: 3:5 30:4 33:5 60:4 91:6 94:5 121:6 124:5 155:6

156: 3:5 4:2 30:4 33:5 59:4 91:6 94:5 121:6 124:5 156:6

173: 7:2 12:3 14:5 19:3 21:5 42:4 103:6 105:5 110:6 112:5 173:6

174: 3:5 4:2 22:3 30:4 33:5 41:4 91:6 94:5 121:6 124:5 174:6

175: 3:5 23:3 30:4 33:5 40:4 91:6 94:5 121:6 124:5 175:6

176: 7:2 12:3 14:5 17:4 19:3 24:5 42:4 103:6 105:5 110:6 115:5 166:6 169:7 176:6

177: 3:5 4:2 8:3 11:5 15:3 18:5 22:3 25:5 30:4 33:5 38:4 91:6 94:5 99:6 102:5 106:6 109:5 113:6 116:5 121:6 124:5 177:6

179: 3:5 11:3 15:4 23:3 30:4 33:5 40:4 48:6 52:4 91:6 94:5 121:6 124:5 163:6 167:7 175:6 179:7

180: 5:3 7:5 12:3 14:5 19:3 21:5 26:3 28:5 35:4 96:6 98:5 103:6 105:5 110:6 112:5 117:6 119:5 180:6

181: 3:5 4:2 29:5 30:4 33:5 34:4 91:6 94:5 95:6 120:5 121:6 124:5 181:6

184: 7:2 31:4 32:6 56:4 159:6 183:7 184:6

185: 21:5 26:3 30:4 51:6 58:4 91:6 112:5 178:6 185:7

188: 2:5 5:4 7:2 30:4 32:6 61:4 91:6 93:5 154:6 157:7 159:6 188:7

194: 7:2 12:3 14:5 19:3 21:5 103:6 105:5 110:6 112:5 194:6

197: 8:5 18:3 55:6 63:4 91:6 99:5 182:6 190:7 197:6

198: 2:4 5:2 28:5 35:4 53:6 61:5 63:4 71:5 96:6 119:5 144:6 162:7 180:6 198:7

199: 2:4 5:2 16:4 22:6 41:4 47:6 58:4 61:5 63:4 66:5 149:6 168:7 174:6 193:7 199:6

200: 26:3 37:4 51:6 73:5 142:6 164:7 178:6 200:7

201: 1:3 12:4 25:3 37:4 50:6 62:5 63:4 141:6 164:7 177:6 190:7 201:6

202: 13:4 25:6 38:4 50:6 63:4 152:6 165:7 177:6 190:7 202:6

203: 12:4 25:6 26:3 37:4 51:6 62:4 152:6 164:7 178:6 189:7 203:6

204: 11:4 15:6 26:3 36:4 51:6 61:5 142:6 163:7 178:6 188:7 204:6

205: 2:4 5:2 21:3 42:4 46:6 61:5 63:4 78:5 137:6 169:7 173:6 205:7

206: 7:2 18:3 19:4 20:6 44:4 45:6 55:5 56:4 79:5 136:6 146:7 147:6 171:7 172:6 206:7

228: 2:4 3:2 4:3 5:2 22:6 23:3 41:5 58:4 61:5 63:4 101:5 114:6 132:7 149:6 228:7

229: 2:4 5:2 22:6 41:5 58:4 61:5 63:4 102:5 113:6 132:7 149:6 229:7

231: 7:2 20:6 43:5 56:4 104:5 111:6 134:7 147:6 231:7

233: 7:2 9:6 18:4 43:5 45:6 70:5 81:6 106:5 109:6 134:7 136:6 170:7 172:6 197:7 208:6 233:7

234: 16:6 36:5 52:6 72:5 88:6 108:5 127:7 143:6 163:7 179:6 199:7 215:6 234:7

235: 7:2 9:6 17:3 43:5 45:6 79:5 81:6 107:5 134:7 136:6 170:7 172:6 206:7 208:6 235:7

236: 7:2 18:6 43:5 45:6 54:4 79:5 81:6 106:5 134:7 145:6 170:7 172:6 206:7 208:6 236:7

238: 7:2 20:6 43:5 56:4 104:5 134:7 147:6 238:7

240: 2:4 5:2 22:6 41:5 58:4 61:5 63:4 102:5 132:7 149:6 240:7

241: 2:4 3:2 4:3 5:2 22:6 23:3 41:5 58:4 61:5 63:4 101:5 132:7 149:6 241:7

243: 25:4 61:5 63:4 86:6 99:5 190:7 213:6 243:7

245: 27:4 61:5 88:6 97:5 188:7 215:6 245:7

246: 2:6 24:3 28:6 36:5 38:4 60:5 61:6 63:4 64:6 96:5 127:7 129:6 151:7 155:6 187:7 188:6 190:7 191:6 246:7

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248: 2:6 5:2 30:6 36:5 38:4 41:5 58:4 61:6 63:4 66:6 94:5 127:7 129:6 132:7 157:6 185:7 188:6 190:7 193:6 248:7

249: 1:3 6:2 31:6 42:5 57:4 62:6 63:4 67:6 93:5 133:7 158:6 184:7 189:6 190:7 194:6 249:7

250: 7:2 32:6 43:5 56:4 68:6 92:5 134:7 159:6 183:7 195:6 250:7

251: 7:2 19:3 33:6 55:5 69:6 91:5 146:7 160:6 182:7 196:6 251:7

252: 7:2 9:6 18:4 34:6 43:5 45:6 54:5 81:6 90:5 134:7 136:6 145:7 161:6 170:7 172:6 181:7 208:6 252:7

253: 2:4 24:3 25:6 27:4 28:6 35:4 60:5 64:6 86:5 88:6 89:5 151:7 152:6 154:7 155:6 187:7 191:6 213:7 215:6 253:7

339: 6:3 20:6 23:4 26:3 172:6 302:8 316:7 322:8 339:7

341: 1:3 21:7 62:6 63:4 189:6 190:7 317:8 341:7

343: 16:5 30:4 46:6 47:4 91:6 107:5 168:6 326:8 342:7 343:8

344: 40:7 42:6 82:5 133:6 167:7 169:8 173:7 175:6 209:7 302:8 342:7 344:8

345: 7:2 166:7 176:8 335:7 345:8

346: 8:5 49:7 63:4 74:6 99:5 165:6 177:8 190:7 201:6 218:7 346:8

347: 9:3 37:4 46:6 51:4 161:6 164:7 330:8 342:7 347:8

348: 10:6 36:5 46:6 52:5 101:6 127:7 137:6 163:7 270:8 342:7 348:8

349: 46:7 53:4 71:6 78:5 162:6 169:8 198:6 205:7 215:6 349:8

350: 9:7 17:4 26:3 54:4 161:6 169:8 178:6 350:8

351: 7:2 19:3 55:7 69:6 97:5 160:6 188:8 196:6 224:7 351:8

353: 6:3 7:2 20:6 25:3 43:5 57:4 116:6 134:7 158:6 285:8 316:7 353:8

354: 5:7 21:4 22:3 58:4 157:6 169:8 173:7 174:6 338:8 342:7 354:8

357: 7:2 19:3 154:7 188:8 323:7 357:8

358: 6:3 7:2 43:7 62:4 85:5 153:6 176:8 189:6 212:7 358:8

363: 4:7 6:6 21:4 42:6 67:5 133:6 148:7 169:8 173:7 300:8 342:7 363:8

B All 191 short alternating-path words

The integer at the left is the new size n; the following one-digit symbols are the successive cube roots r_j in Lemma 13. Together with the explicit seed in Lemma 12, these words form an independently and locally checkable existence certificate for $173 \le n \le 364$.

```
174: 635555363635555363633555536
175: 6336
176: 65555556
177: 6364636
178: 6364625555556
179: 74635555555547
180: 74474247447
181: 6554746474556
182: 655556
183: 663647
184: 66
185: 66264746366
186: 66266
187: 7447
188: 7455665547
189: 666363647447447
190: 647
191: 6665547556
192: 75257
193: 7575555757
194: 66655666
195: 6264757
196: 663666626
197: 75746264757
198: 65575264757
199: 666
200: 66633666
201: 64747574757
202: 6474752556
203: 663466666
204: 757424757
205: 65755756
206: 6666446666
207: 666665546666
208: 64757
209: 6443756
210: 757464757
211: 6666357
212: 66664666
213: 757475757
214: 653756
215: 6666556
216: 7555666666665557
217: 75757375757375757
218: 7555757573573757575557
219: 757455757464757
220: 75757554665575757
221: 757573753666375757
222: 7375757
223: 757455755557554757
224: 7363753637
225: 7557357447557
226: 75447557
227: 757537525735757
228: 7524665566666657
229: 746664667
230: 752757573637557
231: 75666637575725257
232: 757536357556666665757
233: 75666666557537
234: 7375757256666667
235: 77
236: 7466466466466466466466757275747
237: 7375747
238: 74664747
239: 73444667
240: 7466466444667
241: 7337
242: 734664667
243: 73527537534667
244: 73667
245: 746666667
246: 766375375375266667
247: 7666666666666666667
248: 76666266667
249: 766662757475757266667
250: 747
251: 766665736666667
252: 747573375747
253: 7474466375747
254: 74744664664747
255: 74757377
256: 8
257: 76666667
258: 747525747
259: 85666658566277
260: 85662666666667
261: 7475375758
262: 8558
263: 7736666444758
264: 7737574744758
265: 8558534758
266: 766664758
267: 85585757377
268: 7477
269: 766666677477
270: 8575247477
271: 8575266667
272: 7475375375758575758
273: 855857573757447
274: 85266258
275: 7437447
276: 74586666666658
277: 868
278: 7447
279: 868525377
280: 773368
281: 76666477
282: 72755857477
283: 727
284: 777558
285: 727575868
286: 85475758
287: 76658
288: 7577666658
289: 8666857537
290: 8558
291: 856666667
292: 86468
293: 8558667
294: 737575727737
295: 86477737
296: 7775486477
297: 83747775868
298: 7577738
299: 7775468
300: 8646368585486868
301: 7368666468
302: 76644867
303: 734775557
304: 736868
305: 8677777
306: 7586666856768
307: 7684777558
308: 768458667
309: 77776667
310: 76848
311: 77
312: 836366625868
313: 775868577
314: 77457777
315: 7764777
316: 867738
317: 76664468
318: 7668667
319: 864866367
320: 764777
321: 8685485677
322: 7684847777
323: 75737668
324: 7667677367
325: 7586268
326: 7586847
327: 8685758268
328: 8626846668
329: 7577
330: 86856768667
331: 7585268368
332: 867
333: 777777
334: 7686855758
335: 76358668668
336: 784686847586868
337: 8638486777
338: 7468487
339: 77
340: 77583867
341: 7777577
342: 88288
343: 86388
344: 86867777778
345: 858557288
346: 85777488
347: 8586677768
348: 8855676668
349: 8848
350: 8485738686388
351: 86864485882778
352: 8686488
353: 8577664648
354: 85868684848
355: 877388
356: 84577488
357: 86777755778
358: 858557785588
359: 8468
360: 8586482775858
361: 8586855588
362: 883477778
363: 8528686488
364: 855557388
```

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